OUTPUT

The output

This output is very long because my code run for very long time .i have pasted it from google collab directly without editing

---

Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).

Starting appetite prediction pipeline with cross-validation...

Loading data from Google Drive folders...

Files in /content/drive/MyDrive/audio_files/AA:

['audio_files.html', '713_BoAW_openSMILE_2.3.0_eGeMAPS.csv', '687_BoAW_openSMILE_2.3.0_eGeMAPS.csv', '670_BoAW_openSMILE_2.3.0_eGeMAPS.csv', '698_BoAW_openSMILE_2.3.0_eGeMAPS.csv', '632_BoAW_openSMILE_2.3.0_eGeMAPS.csv', '667_BoAW_openSMILE_2.3.0_eGeMAPS.csv', '653_BoAW_openSMILE_2.3.0_eGeMAPS.csv', '657_BoAW_openSMILE_2.3.0_eGeMAPS.csv', '718_BoAW_openSMILE_2.3.0_eGeMAPS.csv'] ...

Files in /content/drive/MyDrive/merged participants:

['merged data of participants', 'Copy of appetite levels mapping.csv', 'appetite levels mapping.csv']

Using mapping file: /content/drive/MyDrive/merged participants/Copy of appetite levels mapping.csv

Mapping file loaded successfully

First few rows of mapping file:

	Participant	PHQ8_5_Appetite	Appetite_Label
0	300	1	loss_or_over
1	301	1	loss_or_over

---

2 302 0 normal

3 303 0 normal

4 304 2 loss_or_over

Columns: ['Participant', 'PHQ8_5_Appetite', 'Appetite_Label']

PHQ8_5_Appetite value counts:

PHQ8_5_Appetite

0 115

1 83

2 44

3 33

Name: count, dtype: int64

Loaded labels for 275 participants

Sample participant labels: [('300', 1), ('301', 1), ('302', 0), ('303', 0), ('304', 2)]

Found 433 CSV files in the participant folder

Found 216 MFCC files and 216 eGeMAPS files

Found 216 complete participant records

Processing participants: 31%|███ | 66/216 [00:08<00:06, 23.50it/s]Skipping participant 718 - empty MFCC data

Processing participants: 100%|██████████| 216/216 [00:16<00:00, 12.84it/s]

Data preprocessing complete

MFCC data shape: (215, 500, 101)

eGeMAPS data shape: (215, 500, 101)

Labels shape: (215,)

Label distribution (rounded): [94 63 34 24]

Processing fold 1/5

Sample weight distribution: (array([0.78991145, 0.90151957, 1.63998653]), array([82, 150, 50]))

Training samples: 282

Test samples: 43

Training label distribution (rounded): [82 50 75 75]

Test label distribution (rounded): [12 13 12 6]

Model architecture:

Model: "functional_5"

```
┏━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━┳━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━
━━┳━━━━━━━━━━━━━━━━━━━━━━━━━┳━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━┓
┃ Layer (type)        ┃ Output Shape       ┃    Param # ┃ Connected to      ┃
┡━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╇━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━
╇━━━━━━━━━━━━━━━━━━━━━━━━━╇━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━┩
│ mfcc_input          │ (None, 500, 101) │          0 │ -                 │
│ (InputLayer)        │                  │            │                   │
├─────────────────────────────┼─────────────────────────┼────────────────┼──────
────────────────────┤
│ egemaps_input        │ (None, 500, 101) │          0 │ -                 │
│ (InputLayer)        │                  │            │                   │
├─────────────────────────────┼─────────────────────────┼────────────────┼──────
────────────────────┤
│ not_equal_4          │ (None, 500, 101) │          0 │ mfcc_input[0][0]  │
│ (NotEqual)          │                  │            │                   │
```

```
├──────────────────────┼──────────────────────┼──────────────┼──────
──────────────────┤
│ not_equal_5        │ (None, 500, 101) │          0 │ egemaps_input[0]… │
│ (NotEqual)         │                  │            │                   │
├──────────────────────┼──────────────────────┼──────────────┼──────
──────────────────┤
│ masking_4 (Masking) │ (None, 500, 101) │          0 │ mfcc_input[0][0]  │
├──────────────────────┼──────────────────────┼──────────────┼──────
──────────────────┤
│ any_4 (Any)        │ (None, 500)      │          0 │ not_equal_4[0][0] │
├──────────────────────┼──────────────────────┼──────────────┼──────
──────────────────┤
│ masking_5 (Masking) │ (None, 500, 101) │          0 │ egemaps_input[0]… │
├──────────────────────┼──────────────────────┼──────────────┼──────
──────────────────┤
│ any_5 (Any)        │ (None, 500)      │          0 │ not_equal_5[0][0] │
├──────────────────────┼──────────────────────┼──────────────┼──────
──────────────────┤
│ lstm_18 (LSTM)     │ (None, 64)       │     42,496 │ masking_4[0][0],  │
│                    │                  │            │ any_4[0][0]       │
├──────────────────────┼──────────────────────┼──────────────┼──────
──────────────────┤
│ lstm_19 (LSTM)     │ (None, 64)       │     42,496 │ masking_5[0][0],  │
│                    │                  │            │ any_5[0][0]       │
├──────────────────────┼──────────────────────┼──────────────┼──────
──────────────────┤
│ batch_normalizatio… │ (None, 64)       │        256 │ lstm_18[0][0]     │
```

(BatchNormalizatio…			
batch_normalizatio…	(None, 64)	256	lstm_19[0][0]
(BatchNormalizatio…			
dropout_8 (Dropout)	(None, 64)	0	batch_normalizat…
dropout_9 (Dropout)	(None, 64)	0	batch_normalizat…
concatenate_5	(None, 128)	0	dropout_8[0][0],
(Concatenate)			dropout_9[0][0]
dense_6 (Dense)	(None, 32)	4,128	concatenate_5[0]…
dropout_10	(None, 32)	0	dense_6[0][0]
(Dropout)			
dense_7 (Dense)	(None, 1)	33	dropout_10[0][0]

Total params: 89,665 (350.25 KB)

Trainable params: 89,409 (349.25 KB)

Non-trainable params: 256 (1.00 KB)

Epoch 1/50

15/15 ━━━━━━━━━━━━━━━━━━━━ 23s 835ms/step - loss: 1.4101 - mae: 1.8750 - val_loss: 0.9297 - val_mae: 1.3902 - learning_rate: 0.0010

Epoch 2/50

15/15 ━━━━━━━━━━━━━━━━━━━━ 20s 820ms/step - loss: 1.1258 - mae: 1.5644 - val_loss: 0.6970 - val_mae: 1.1612 - learning_rate: 0.0010

Epoch 3/50

15/15 ━━━━━━━━━━━━━━━━━━━━ 19s 677ms/step - loss: 0.8261 - mae: 1.2990 - val_loss: 0.5943 - val_mae: 1.0614 - learning_rate: 0.0010

Epoch 4/50

15/15 ━━━━━━━━━━━━━━━━━━━━ 12s 824ms/step - loss: 0.6006 - mae: 1.0333 - val_loss: 0.5763 - val_mae: 1.0428 - learning_rate: 0.0010

Epoch 5/50

15/15 ━━━━━━━━━━━━━━━━━━━━ 18s 646ms/step - loss: 0.7918 - mae: 1.2332 - val_loss: 0.4935 - val_mae: 0.9484 - learning_rate: 0.0010

Epoch 6/50

15/15 ━━━━━━━━━━━━━━━━━━━━ 12s 807ms/step - loss: 0.6948 - mae: 1.1211 - val_loss: 0.4634 - val_mae: 0.9127 - learning_rate: 0.0010

Epoch 7/50

15/15 ━━━━━━━━━━━━━━━━━━━━ 18s 638ms/step - loss: 0.5265 - mae: 0.9470 - val_loss: 0.4555 - val_mae: 0.9061 - learning_rate: 0.0010

Epoch 8/50

15/15 ━━━━━━━━━━━━━━━━━━━━ 12s 807ms/step - loss: 0.5533 - mae: 0.9547 - val_loss: 0.4244 - val_mae: 0.8684 - learning_rate: 0.0010

Epoch 9/50

15/15 ━━━━━━━━━━━━━━━━━━━━ 18s 629ms/step - loss: 0.5331 - mae: 0.9489 - val_loss: 0.3232 - val_mae: 0.7527 - learning_rate: 0.0010

Epoch 10/50

15/15 ━━━━━━━━━━━━━━━━━━━━ 12s 732ms/step - loss: 0.4084 - mae: 0.7768 - val_loss: 0.2960 - val_mae: 0.7134 - learning_rate: 0.0010

Epoch 11/50

15/15 ━━━━━━━━━━━━━━━━━━━━ 20s 760ms/step - loss: 0.4385 - mae: 0.8543 - val_loss: 0.2781 - val_mae: 0.6900 - learning_rate: 0.0010

Epoch 12/50

15/15 ━━━━━━━━━━━━━━━━━━━━ 21s 820ms/step - loss: 0.3812 - mae: 0.7651 - val_loss: 0.2891 - val_mae: 0.6893 - learning_rate: 0.0010

Epoch 13/50

15/15 ━━━━━━━━━━━━━━━━━━━━ 18s 625ms/step - loss: 0.4228 - mae: 0.8142 - val_loss: 0.2642 - val_mae: 0.6553 - learning_rate: 0.0010

Epoch 14/50

15/15 ━━━━━━━━━━━━━━━━━━━━ 12s 809ms/step - loss: 0.4020 - mae: 0.7612 - val_loss: 0.2173 - val_mae: 0.6015 - learning_rate: 0.0010

Epoch 15/50

15/15 ━━━━━━━━━━━━━━━━━━━━ 18s 617ms/step - loss: 0.4308 - mae: 0.8365 - val_loss: 0.2726 - val_mae: 0.6492 - learning_rate: 0.0010

Epoch 16/50

15/15 ━━━━━━━━━━━━━━━━━━━━ 12s 807ms/step - loss: 0.3869 - mae: 0.7807 - val_loss: 0.2440 - val_mae: 0.6143 - learning_rate: 0.0010

Epoch 17/50

15/15 ━━━━━━━━━━━━━━━━━━━━ 18s 627ms/step - loss: 0.3877 - mae: 0.7872 - val_loss: 0.2338 - val_mae: 0.5954 - learning_rate: 0.0010

Epoch 18/50

15/15 ━━━━━━━━━━━━━━━━━━━━ 12s 726ms/step - loss: 0.3393 - mae: 0.7145 - val_loss: 0.2138 - val_mae: 0.5677 - learning_rate: 5.0000e-04

Epoch 19/50

15/15 ━━━━━━━━━━━━━━━━━━━━ 20s 764ms/step - loss: 0.3407 - mae: 0.7144 - val_loss: 0.2180 - val_mae: 0.5619 - learning_rate: 5.0000e-04

Epoch 20/50

15/15 ━━━━━━━━━━━━━━━━━━━━ 21s 817ms/step - loss: 0.3384 - mae: 0.6887 - val_loss: 0.2171 - val_mae: 0.5546 - learning_rate: 5.0000e-04

Epoch 21/50

15/15 ━━━━━━━━━━━━━━━━━━━━ 12s 799ms/step - loss: 0.3671 - mae: 0.7471 - val_loss: 0.2151 - val_mae: 0.5355 - learning_rate: 5.0000e-04

Epoch 22/50

15/15 ━━━━━━━━━━━━━━━━━━━━ 9s 616ms/step - loss: 0.3578 - mae: 0.7045 - val_loss: 0.2158 - val_mae: 0.5313 - learning_rate: 2.5000e-04

Epoch 23/50

15/15 ━━━━━━━━━━━━━━━━━━━━ 13s 755ms/step - loss: 0.2586 - mae: 0.6219 - val_loss: 0.2139 - val_mae: 0.5286 - learning_rate: 2.5000e-04

WARNING:tensorflow:5 out of the last 5 calls to <function TensorFlowTrainer.make_predict_function.<locals>.one_step_on_data_distributed at 0x7f5cacbde5c0> triggered tf.function retracing. Tracing is expensive and the excessive number of tracings could be due to (1) creating @tf.function repeatedly in a loop, (2) passing tensors with different shapes, (3) passing Python objects instead of tensors. For (1), please define your @tf.function outside of the loop. For (2), @tf.function has reduce_retracing=True option that can avoid unnecessary retracing. For (3), please refer to https://www.tensorflow.org/guide/function#controlling_retracing and https://www.tensorflow.org/api_docs/python/tf/function for more details.

WARNING:tensorflow:6 out of the last 6 calls to <function TensorFlowTrainer.make_predict_function.<locals>.one_step_on_data_distributed at 0x7f5cacbde5c0> triggered tf.function retracing. Tracing is expensive and the excessive number of tracings could be due to (1) creating @tf.function repeatedly in a loop, (2) passing tensors with different shapes, (3) passing Python objects instead of tensors. For (1), please define your @tf.function outside of the loop. For (2), @tf.function has reduce_retracing=True option that can avoid unnecessary retracing. For (3), please refer to https://www.tensorflow.org/guide/function#controlling_retracing and https://www.tensorflow.org/api_docs/python/tf/function for more details.

Fold 1 results:

Test MSE: 0.6518

Test MAE: 1.0445

Test R^2: -0.6099

Per-bin evaluation:

Bin 0 (n=12): MSE=0.6553, MAE=0.7572

Bin 1 (n=13): MSE=0.1506, MAE=0.2827

Bin 2 (n=12): MSE=2.3594, MAE=1.5113

Bin 3 (n=6): MSE=5.6252, MAE=2.3358

Predictions saved to /content/drive/MyDrive/test_predictions_regression_fold_1.csv

Sample predictions:

True: 1.00, Predicted: 1.08

True: 2.00, Predicted: 0.36

True: 0.00, Predicted: 0.61

True: 2.00, Predicted: 0.74

True: 3.00, Predicted: 0.64

True: 2.00, Predicted: 0.62

True: 0.00, Predicted: 0.94

True: 0.00, Predicted: 1.20

True: 0.00, Predicted: 1.11

True: 1.00, Predicted: 1.24

Processing fold 2/5

Sample weight distribution: (array([0.85925055, 1.65976304]), array([225, 48]))

Training samples: 273

Test samples: 43

Training label distribution (rounded): [75 48 75 75]

Test label distribution (rounded): [19 15 5 4]

Model architecture:

Model: "functional_6"

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┏━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━┳━━━━━━━━━━━━━━━━━━━━━━━━━━━━━
━━┳━━━━━━━━━━━━━━━━━━━┳━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━┓
┃ Layer (type)        ┃ Output Shape       ┃    Param # ┃ Connected to      ┃
┡━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╇━━━━━━━━━━━━━━━━━━━━━━━━━━━━━
╇━━━━━━━━━━━━━━━━━━━╇━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━┩
│ mfcc_input          │ (None, 500, 101) │          0 │ -                 │
│ (InputLayer)        │                  │            │                   │
├───────────────────────────┼────────────────────────┼──────────────┼──────
─────────────────┤
│ egemaps_input       │ (None, 500, 101) │          0 │ -                 │
```

```
│ (InputLayer)      │                   │            │                      │
├───────────────────┼───────────────────┼────────────┼──────
──────────────┤
│ not_equal_6       │ (None, 500, 101)  │          0 │ mfcc_input[0][0]     │
│ (NotEqual)        │                   │            │                      │
├───────────────────┼───────────────────┼────────────┼──────
──────────────┤
│ not_equal_7       │ (None, 500, 101)  │          0 │ egemaps_input[0]…    │
│ (NotEqual)        │                   │            │                      │
├───────────────────┼───────────────────┼────────────┼──────
──────────────┤
│ masking_6 (Masking) │ (None, 500, 101) │          0 │ mfcc_input[0][0]   │
├───────────────────┼───────────────────┼────────────┼──────
──────────────┤
│ any_6 (Any)       │ (None, 500)       │          0 │ not_equal_6[0][0]    │
├───────────────────┼───────────────────┼────────────┼──────
──────────────┤
│ masking_7 (Masking) │ (None, 500, 101) │          0 │ egemaps_input[0]…  │
├───────────────────┼───────────────────┼────────────┼──────
──────────────┤
│ any_7 (Any)       │ (None, 500)       │          0 │ not_equal_7[0][0]    │
├───────────────────┼───────────────────┼────────────┼──────
──────────────┤
│ lstm_20 (LSTM)    │ (None, 64)        │     42,496 │ masking_6[0][0],     │
│                   │                   │            │ any_6[0][0]          │
├───────────────────┼───────────────────┼────────────┼──────
──────────────┤
```

lstm_21 (LSTM)	(None, 64)	42,496	masking_7[0][0], any_7[0][0]
batch_normalizatio… (BatchNormalizatio…	(None, 64)	256	lstm_20[0][0]
batch_normalizatio… (BatchNormalizatio…	(None, 64)	256	lstm_21[0][0]
dropout_11 (Dropout)	(None, 64)	0	batch_normalizat…
dropout_12 (Dropout)	(None, 64)	0	batch_normalizat…
concatenate_6 (Concatenate)	(None, 128)	0	dropout_11[0][0], dropout_12[0][0]
dense_8 (Dense)	(None, 32)	4,128	concatenate_6[0]…

```
│ dropout_13          │ (None, 32)          │          0 │ dense_8[0][0]     │
│ (Dropout)           │                     │            │                   │
├─────────────────────┼─────────────────────┼────────────┼───────────────────┤
│ dense_9 (Dense)     │ (None, 1)           │         33 │ dropout_13[0][0]  │
└─────────────────────┴─────────────────────┴────────────┴───────────────────┘
 Total params: 89,665 (350.25 KB)
 Trainable params: 89,409 (349.25 KB)
 Non-trainable params: 256 (1.00 KB)
Epoch 1/50
14/14 ━━━━━━━━━━━━━━━━━━━━ 22s 884ms/step - loss: 2.6882 - mae: 3.3049 - val_loss: 1.1459 - val_mae: 1.6673 - learning_rate: 0.0010
Epoch 2/50
14/14 ━━━━━━━━━━━━━━━━━━━━ 9s 620ms/step - loss: 1.9328 - mae: 2.4670 - val_loss: 0.9549 - val_mae: 1.4466 - learning_rate: 0.0010
Epoch 3/50
14/14 ━━━━━━━━━━━━━━━━━━━━ 12s 685ms/step - loss: 1.3292 - mae: 1.8788 - val_loss: 0.7216 - val_mae: 1.2179 - learning_rate: 0.0010
Epoch 4/50
14/14 ━━━━━━━━━━━━━━━━━━━━ 11s 810ms/step - loss: 1.2025 - mae: 1.6564 - val_loss: 0.6115 - val_mae: 1.1057 - learning_rate: 0.0010
Epoch 5/50
14/14 ━━━━━━━━━━━━━━━━━━━━ 9s 663ms/step - loss: 0.9170 - mae: 1.3404 - val_loss: 0.5708 - val_mae: 1.0584 - learning_rate: 0.0010
Epoch 6/50
```

14/14 ━━━━━━━━━━━━━━━━━━━━ 10s 629ms/step - loss: 0.9217 - mae: 1.3813 - val_loss: 0.5461 - val_mae: 1.0310 - learning_rate: 0.0010

Epoch 7/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 11s 814ms/step - loss: 0.8368 - mae: 1.2425 - val_loss: 0.4937 - val_mae: 0.9765 - learning_rate: 0.0010

Epoch 8/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 19s 661ms/step - loss: 0.5882 - mae: 1.0112 - val_loss: 0.4640 - val_mae: 0.9378 - learning_rate: 0.0010

Epoch 9/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 12s 819ms/step - loss: 0.6384 - mae: 1.0289 - val_loss: 0.4513 - val_mae: 0.9054 - learning_rate: 0.0010

Epoch 10/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 19s 662ms/step - loss: 0.7428 - mae: 1.1777 - val_loss: 0.4087 - val_mae: 0.8543 - learning_rate: 0.0010

Epoch 11/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 11s 825ms/step - loss: 0.6994 - mae: 1.1114 - val_loss: 0.4007 - val_mae: 0.8446 - learning_rate: 0.0010

Epoch 12/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 19s 678ms/step - loss: 0.6186 - mae: 1.0169 - val_loss: 0.3963 - val_mae: 0.8460 - learning_rate: 0.0010

Epoch 13/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 12s 818ms/step - loss: 0.5848 - mae: 0.9993 - val_loss: 0.3893 - val_mae: 0.8152 - learning_rate: 0.0010

Epoch 14/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 19s 696ms/step - loss: 0.6229 - mae: 1.0112 - val_loss: 0.3741 - val_mae: 0.7922 - learning_rate: 0.0010

Epoch 15/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 11s 820ms/step - loss: 0.5016 - mae: 0.9012 - val_loss: 0.3593 - val_mae: 0.7861 - learning_rate: 0.0010

Epoch 16/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 9s 633ms/step - loss: 0.5608 - mae: 1.0097 - val_loss: 0.3554 - val_mae: 0.7630 - learning_rate: 0.0010

Epoch 17/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 11s 675ms/step - loss: 0.4986 - mae: 0.8924 - val_loss: 0.3307 - val_mae: 0.7163 - learning_rate: 0.0010

Epoch 18/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 11s 807ms/step - loss: 0.4684 - mae: 0.8735 - val_loss: 0.3032 - val_mae: 0.6820 - learning_rate: 0.0010

Epoch 19/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 9s 676ms/step - loss: 0.5117 - mae: 0.9416 - val_loss: 0.3028 - val_mae: 0.6856 - learning_rate: 0.0010

Epoch 20/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 11s 685ms/step - loss: 0.4888 - mae: 0.8746 - val_loss: 0.3009 - val_mae: 0.6648 - learning_rate: 0.0010

Epoch 21/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 11s 814ms/step - loss: 0.3800 - mae: 0.7670 - val_loss: 0.2683 - val_mae: 0.6356 - learning_rate: 0.0010

Epoch 22/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 19s 687ms/step - loss: 0.3159 - mae: 0.6836 - val_loss: 0.2916 - val_mae: 0.6729 - learning_rate: 0.0010

Epoch 23/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 11s 755ms/step - loss: 0.3398 - mae: 0.7146 - val_loss: 0.2695 - val_mae: 0.6635 - learning_rate: 0.0010

Epoch 24/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 19s 620ms/step - loss: 0.3738 - mae: 0.7546 - val_loss: 0.2748 - val_mae: 0.6531 - learning_rate: 0.0010

Epoch 25/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 11s 782ms/step - loss: 0.3225 - mae: 0.6812 - val_loss: 0.2781 - val_mae: 0.6378 - learning_rate: 5.0000e-04

Epoch 26/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 20s 715ms/step - loss: 0.3005 - mae: 0.6726 - val_loss: 0.2694 - val_mae: 0.6179 - learning_rate: 5.0000e-04

Fold 2 results:

Test MSE: 0.4649

Test MAE: 0.8647

Test R^2: -0.2223

Per-bin evaluation:

Bin 0 (n=19): MSE=0.8662, MAE=0.8471

Bin 1 (n=15): MSE=0.3069, MAE=0.4492

Bin 2 (n=5): MSE=1.5566, MAE=1.2272

Bin 3 (n=4): MSE=4.7562, MAE=2.0540

Predictions saved to /content/drive/MyDrive/test_predictions_regression_fold_2.csv

Sample predictions:

True: 0.00, Predicted: 0.79

True: 3.00, Predicted: 1.69

True: 2.00, Predicted: 0.45

True: 0.00, Predicted: 0.68

True: 1.00, Predicted: 0.46

True: 3.00, Predicted: -0.20

True: 1.00, Predicted: 0.41

True: 1.00, Predicted: 0.61

True: 1.00, Predicted: 0.55

True: 3.00, Predicted: 0.84

Processing fold 3/5

Sample weight distribution: (array([0.85774462, 0.93020709, 1.51556084]), array([150, 71, 51]))

Training samples: 272

Test samples: 43

Training label distribution (rounded): [71 51 75 75]

Test label distribution (rounded): [23 12 5 3]

Model architecture:

Model: "functional_7"

```
┏━━━━━━━━━━━━━━━━━━━━━━━┳━━━━━━━━━━━━━━━━━━━━━━┳━━━━━━━━━━━━━┳━━━━━━━━━━━━━━━━━━━━━━━┓
┃ Layer (type)        ┃ Output Shape      ┃    Param # ┃ Connected to      ┃
┡━━━━━━━━━━━━━━━━━━━━━━━╇━━━━━━━━━━━━━━━━━━━━━━╇━━━━━━━━━━━━━╇━━━━━━━━━━━━━━━━━━━━━━━┩
│ mfcc_input        │ (None, 500, 101) │        0 │ -              │
│ (InputLayer)       │             │         │              │
├───────────────────────┼──────────────────────┼─────────────┼───────────────────────┤
```

```
│ egemaps_input       │ (None, 500, 101) │          0 │ -                 │
│ (InputLayer)        │                  │            │                   │
├─────────────────────┼──────────────────┼────────────┼───────
──────────────────┤
│ not_equal_8         │ (None, 500, 101) │          0 │ mfcc_input[0][0]  │
│ (NotEqual)          │                  │            │                   │
├─────────────────────┼──────────────────┼────────────┼───────
──────────────────┤
│ not_equal_9         │ (None, 500, 101) │          0 │ egemaps_input[0]… │
│ (NotEqual)          │                  │            │                   │
├─────────────────────┼──────────────────┼────────────┼───────
──────────────────┤
│ masking_8 (Masking) │ (None, 500, 101) │          0 │ mfcc_input[0][0]  │
├─────────────────────┼──────────────────┼────────────┼───────
──────────────────┤
│ any_8 (Any)         │ (None, 500)      │          0 │ not_equal_8[0][0] │
├─────────────────────┼──────────────────┼────────────┼───────
──────────────────┤
│ masking_9 (Masking) │ (None, 500, 101) │          0 │ egemaps_input[0]… │
├─────────────────────┼──────────────────┼────────────┼───────
──────────────────┤
│ any_9 (Any)         │ (None, 500)      │          0 │ not_equal_9[0][0] │
├─────────────────────┼──────────────────┼────────────┼───────
──────────────────┤
│ lstm_22 (LSTM)      │ (None, 64)       │     42,496 │ masking_8[0][0],  │
│                     │                  │            │ any_8[0][0]       │
```

lstm_23 (LSTM)	(None, 64)	42,496	masking_9[0][0], any_9[0][0]
batch_normalizatio… (BatchNormalizatio…	(None, 64)	256	lstm_22[0][0]
batch_normalizatio… (BatchNormalizatio…	(None, 64)	256	lstm_23[0][0]
dropout_14 (Dropout)	(None, 64)	0	batch_normalizat…
dropout_15 (Dropout)	(None, 64)	0	batch_normalizat…
concatenate_7 (Concatenate)	(None, 128)	0	dropout_14[0][0], dropout_15[0][0]
dense_10 (Dense)	(None, 32)	4,128	concatenate_7[0]…

```
├──────────────────────────────┼───────────────────────────┼─────────────────┼──────
─────────────────────┤
│ dropout_16          │ (None, 32)          │          0 │ dense_10[0][0]      │
│ (Dropout)           │                     │            │                     │
├──────────────────────────────┼───────────────────────────┼─────────────────┼──────
─────────────────────┤
│ dense_11 (Dense)    │ (None, 1)           │         33 │ dropout_16[0][0]    │
└──────────────────────────────┴───────────────────────────┴─────────────────┴──────
─────────────────────┘
 Total params: 89,665 (350.25 KB)
 Trainable params: 89,409 (349.25 KB)
 Non-trainable params: 256 (1.00 KB)
Epoch 1/50
14/14 ━━━━━━━━━━━━━━━━━━━━ 20s 862ms/step - loss: 1.0785 - mae:
1.6024 - val_loss: 0.6533 - val_mae: 1.1494 - learning_rate: 0.0010
Epoch 2/50
14/14 ━━━━━━━━━━━━━━━━━━━━ 20s 777ms/step - loss: 0.8202 - mae:
1.2626 - val_loss: 0.5875 - val_mae: 1.0717 - learning_rate: 0.0010
Epoch 3/50
14/14 ━━━━━━━━━━━━━━━━━━━━ 11s 786ms/step - loss: 0.5886 - mae:
0.9911 - val_loss: 0.5296 - val_mae: 0.9956 - learning_rate: 0.0010
Epoch 4/50
14/14 ━━━━━━━━━━━━━━━━━━━━ 20s 743ms/step - loss: 0.6128 - mae:
1.0556 - val_loss: 0.4550 - val_mae: 0.9146 - learning_rate: 0.0010
Epoch 5/50
14/14 ━━━━━━━━━━━━━━━━━━━━ 18s 585ms/step - loss: 0.6107 - mae:
1.0444 - val_loss: 0.4326 - val_mae: 0.8854 - learning_rate: 0.0010
```

Epoch 6/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 11s 616ms/step - loss: 0.5817 - mae: 1.0204 - val_loss: 0.3915 - val_mae: 0.8330 - learning_rate: 0.0010

Epoch 7/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 11s 766ms/step - loss: 0.6301 - mae: 1.0633 - val_loss: 0.3945 - val_mae: 0.8327 - learning_rate: 0.0010

Epoch 8/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 19s 616ms/step - loss: 0.4900 - mae: 0.8896 - val_loss: 0.3848 - val_mae: 0.8366 - learning_rate: 0.0010

Epoch 9/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 12s 774ms/step - loss: 0.4116 - mae: 0.8334 - val_loss: 0.3297 - val_mae: 0.7680 - learning_rate: 0.0010

Epoch 10/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 9s 662ms/step - loss: 0.5134 - mae: 0.9194 - val_loss: 0.3264 - val_mae: 0.7617 - learning_rate: 0.0010

Epoch 11/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 10s 623ms/step - loss: 0.3948 - mae: 0.7543 - val_loss: 0.3152 - val_mae: 0.7455 - learning_rate: 0.0010

Epoch 12/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 11s 822ms/step - loss: 0.3393 - mae: 0.7311 - val_loss: 0.3248 - val_mae: 0.7611 - learning_rate: 0.0010

Epoch 13/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 10s 721ms/step - loss: 0.4370 - mae: 0.8069 - val_loss: 0.3288 - val_mae: 0.7596 - learning_rate: 0.0010

Epoch 14/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 10s 676ms/step - loss: 0.4619 - mae: 0.8281 - val_loss: 0.2622 - val_mae: 0.6590 - learning_rate: 0.0010

Epoch 15/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 12s 836ms/step - loss: 0.3469 - mae: 0.7205 - val_loss: 0.2289 - val_mae: 0.6148 - learning_rate: 0.0010

Epoch 16/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 19s 659ms/step - loss: 0.3493 - mae: 0.7301 - val_loss: 0.2215 - val_mae: 0.6002 - learning_rate: 0.0010

Epoch 17/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 12s 831ms/step - loss: 0.3433 - mae: 0.7250 - val_loss: 0.1945 - val_mae: 0.5495 - learning_rate: 0.0010

Epoch 18/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 13s 978ms/step - loss: 0.2612 - mae: 0.6111 - val_loss: 0.2021 - val_mae: 0.5515 - learning_rate: 0.0010

Epoch 19/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 13s 857ms/step - loss: 0.2916 - mae: 0.6514 - val_loss: 0.2017 - val_mae: 0.5656 - learning_rate: 0.0010

Epoch 20/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 12s 901ms/step - loss: 0.3345 - mae: 0.7181 - val_loss: 0.1877 - val_mae: 0.5322 - learning_rate: 0.0010

Epoch 21/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 10s 746ms/step - loss: 0.2417 - mae: 0.5649 - val_loss: 0.1973 - val_mae: 0.5469 - learning_rate: 0.0010

Epoch 22/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 21s 824ms/step - loss: 0.2979 - mae: 0.6383 - val_loss: 0.1908 - val_mae: 0.5507 - learning_rate: 0.0010

Epoch 23/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 18s 639ms/step - loss: 0.2513 - mae: 0.6086 - val_loss: 0.1755 - val_mae: 0.4949 - learning_rate: 0.0010

Epoch 24/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 11s 824ms/step - loss: 0.2396 - mae: 0.5885 - val_loss: 0.1758 - val_mae: 0.5011 - learning_rate: 0.0010

Epoch 25/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 18s 626ms/step - loss: 0.2669 - mae: 0.6175 - val_loss: 0.1562 - val_mae: 0.4635 - learning_rate: 0.0010

Epoch 26/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 13s 825ms/step - loss: 0.2134 - mae: 0.5593 - val_loss: 0.1507 - val_mae: 0.4413 - learning_rate: 0.0010

Epoch 27/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 12s 875ms/step - loss: 0.1961 - mae: 0.5225 - val_loss: 0.1655 - val_mae: 0.4804 - learning_rate: 0.0010

Epoch 28/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 20s 830ms/step - loss: 0.2331 - mae: 0.5719 - val_loss: 0.1764 - val_mae: 0.4998 - learning_rate: 0.0010

Epoch 29/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 11s 826ms/step - loss: 0.2014 - mae: 0.5342 - val_loss: 0.1637 - val_mae: 0.4597 - learning_rate: 0.0010

Epoch 30/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 9s 635ms/step - loss: 0.2402 - mae: 0.5898 - val_loss: 0.1644 - val_mae: 0.4708 - learning_rate: 5.0000e-04

Epoch 31/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 13s 787ms/step - loss: 0.2225 - mae: 0.5382 - val_loss: 0.1594 - val_mae: 0.4591 - learning_rate: 5.0000e-04

Fold 3 results:

Test MSE: 0.5561

Test MAE: 0.9681

Test R^2: -0.5753

Per-bin evaluation:

Bin 0 (n=23): MSE=1.2001, MAE=0.9962

Bin 1 (n=12): MSE=0.2239, MAE=0.3943

Bin 2 (n=5): MSE=1.9816, MAE=1.3681

Bin 3 (n=3): MSE=5.8462, MAE=2.3813

Predictions saved to /content/drive/MyDrive/test_predictions_regression_fold_3.csv

Sample predictions:

True: 0.00, Predicted: 1.03

True: 1.00, Predicted: 0.69

True: 0.00, Predicted: 0.94

True: 1.00, Predicted: 0.14

True: 0.00, Predicted: 0.71

True: 0.00, Predicted: 1.32

True: 1.00, Predicted: 0.84

True: 1.00, Predicted: 0.55

True: 2.00, Predicted: 0.53

True: 0.00, Predicted: 0.04

Processing fold 4/5

Sample weight distribution: (array([0.85587215, 0.89080306, 1.60388887]), array([150, 73, 49]))

Training samples: 272

Test samples: 43

Training label distribution (rounded): [73 49 75 75]

Test label distribution (rounded): [21 14 3 5]

Model architecture:

Model: "functional_8"

```
┏━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━┳━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━
━━┳━━━━━━━━━━━━━━━━━━━━━━━┳━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━┓
┃ Layer (type)        ┃ Output Shape       ┃    Param # ┃ Connected to      ┃
┡━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╇━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━
╇━━━━━━━━━━━━━━━━━━━━━━━╇━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━┩
│ mfcc_input          │ (None, 500, 101) │          0 │ -                 │
│ (InputLayer)        │                  │            │                   │
├─────────────────────┼──────────────────┼────────────┼───────
───────────────────┤
│ egemaps_input       │ (None, 500, 101) │          0 │ -                 │
│ (InputLayer)        │                  │            │                   │
├─────────────────────┼──────────────────┼────────────┼───────
───────────────────┤
│ not_equal_10        │ (None, 500, 101) │          0 │ mfcc_input[0][0]  │
│ (NotEqual)          │                  │            │                   │
├─────────────────────┼──────────────────┼────────────┼───────
───────────────────┤
│ not_equal_11        │ (None, 500, 101) │          0 │ egemaps_input[0]… │
│ (NotEqual)          │                  │            │                   │
```

```
├───────────────────────┼─────────────────────┼──────────────┼─────
─────────────────┤
│ masking_10          │ (None, 500, 101) │          0 │ mfcc_input[0][0]  │
│ (Masking)           │                   │            │                   │
├───────────────────────┼─────────────────────┼──────────────┼─────
─────────────────┤
│ any_10 (Any)        │ (None, 500)       │          0 │ not_equal_10[0][… │
├───────────────────────┼─────────────────────┼──────────────┼─────
─────────────────┤
│ masking_11          │ (None, 500, 101) │          0 │ egemaps_input[0]… │
│ (Masking)           │                   │            │                   │
├───────────────────────┼─────────────────────┼──────────────┼─────
─────────────────┤
│ any_11 (Any)        │ (None, 500)       │          0 │ not_equal_11[0][… │
├───────────────────────┼─────────────────────┼──────────────┼─────
─────────────────┤
│ lstm_24 (LSTM)      │ (None, 64)        │     42,496 │ masking_10[0][0], │
│                     │                   │            │ any_10[0][0]      │
├───────────────────────┼─────────────────────┼──────────────┼─────
─────────────────┤
│ lstm_25 (LSTM)      │ (None, 64)        │     42,496 │ masking_11[0][0], │
│                     │                   │            │ any_11[0][0]      │
├───────────────────────┼─────────────────────┼──────────────┼─────
─────────────────┤
│ batch_normalizatio… │ (None, 64)        │        256 │ lstm_24[0][0]     │
│ (BatchNormalizatio… │                   │            │                   │
```

```
├───────────────────────────┼────────────────────────┼───────────────┼──────
──────────────────┤
│ batch_normalizatio… │ (None, 64)      │     256 │ lstm_25[0][0]    │
│ (BatchNormalizatio… │                 │         │                  │
├───────────────────────────┼────────────────────────┼───────────────┼──────
──────────────────┤
│ dropout_17          │ (None, 64)      │       0 │ batch_normalizat… │
│ (Dropout)           │                 │         │                  │
├───────────────────────────┼────────────────────────┼───────────────┼──────
──────────────────┤
│ dropout_18          │ (None, 64)      │       0 │ batch_normalizat… │
│ (Dropout)           │                 │         │                  │
├───────────────────────────┼────────────────────────┼───────────────┼──────
──────────────────┤
│ concatenate_8       │ (None, 128)     │       0 │ dropout_17[0][0], │
│ (Concatenate)       │                 │         │ dropout_18[0][0]  │
├───────────────────────────┼────────────────────────┼───────────────┼──────
──────────────────┤
│ dense_12 (Dense)    │ (None, 32)      │   4,128 │ concatenate_8[0]… │
├───────────────────────────┼────────────────────────┼───────────────┼──────
──────────────────┤
│ dropout_19          │ (None, 32)      │       0 │ dense_12[0][0]    │
│ (Dropout)           │                 │         │                  │
├───────────────────────────┼────────────────────────┼───────────────┼──────
──────────────────┤
│ dense_13 (Dense)    │ (None, 1)       │      33 │ dropout_19[0][0]  │
```

└─────────────────────────────┴──────────────────────────┴─────────────────┴──────
─────────────────────┘

```
 Total params: 89,665 (350.25 KB)

 Trainable params: 89,409 (349.25 KB)

 Non-trainable params: 256 (1.00 KB)

Epoch 1/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 23s 916ms/step - loss: 1.3874 - mae: 1.9094 - val_loss: 0.7115 - val_mae: 1.2185 - learning_rate: 0.0010

Epoch 2/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 17s 641ms/step - loss: 0.9247 - mae: 1.3476 - val_loss: 0.6901 - val_mae: 1.1823 - learning_rate: 0.0010

Epoch 3/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 12s 858ms/step - loss: 0.6785 - mae: 1.1095 - val_loss: 0.6615 - val_mae: 1.1418 - learning_rate: 0.0010

Epoch 4/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 11s 835ms/step - loss: 0.5809 - mae: 0.9880 - val_loss: 0.6360 - val_mae: 1.1046 - learning_rate: 0.0010

Epoch 5/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 21s 849ms/step - loss: 0.5942 - mae: 0.9609 - val_loss: 0.5622 - val_mae: 1.0266 - learning_rate: 0.0010

Epoch 6/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 18s 637ms/step - loss: 0.6413 - mae: 1.0715 - val_loss: 0.5284 - val_mae: 0.9881 - learning_rate: 0.0010

Epoch 7/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 12s 857ms/step - loss: 0.5462 - mae: 0.9934 - val_loss: 0.5541 - val_mae: 1.0079 - learning_rate: 0.0010

Epoch 8/50
```

14/14 ━━━━━━━━━━━━━━━━━━━━ 18s 635ms/step - loss: 0.4355 - mae: 0.8144 - val_loss: 0.5584 - val_mae: 1.0281 - learning_rate: 0.0010

Epoch 9/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 13s 770ms/step - loss: 0.4355 - mae: 0.8482 - val_loss: 0.5023 - val_mae: 0.9728 - learning_rate: 0.0010

Epoch 10/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 20s 756ms/step - loss: 0.4398 - mae: 0.8279 - val_loss: 0.4808 - val_mae: 0.9462 - learning_rate: 0.0010

Epoch 11/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 22s 849ms/step - loss: 0.3799 - mae: 0.7412 - val_loss: 0.4867 - val_mae: 0.9426 - learning_rate: 0.0010

Epoch 12/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 18s 646ms/step - loss: 0.3874 - mae: 0.7998 - val_loss: 0.4746 - val_mae: 0.9243 - learning_rate: 0.0010

Epoch 13/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 13s 828ms/step - loss: 0.3890 - mae: 0.7767 - val_loss: 0.3866 - val_mae: 0.8240 - learning_rate: 0.0010

Epoch 14/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 12s 852ms/step - loss: 0.3959 - mae: 0.7422 - val_loss: 0.3794 - val_mae: 0.8235 - learning_rate: 0.0010

Epoch 15/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 21s 830ms/step - loss: 0.3815 - mae: 0.7611 - val_loss: 0.4148 - val_mae: 0.8718 - learning_rate: 0.0010

Epoch 16/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 12s 858ms/step - loss: 0.4165 - mae: 0.8301 - val_loss: 0.3467 - val_mae: 0.7847 - learning_rate: 0.0010

Epoch 17/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 9s 675ms/step - loss: 0.3132 - mae: 0.6819 - val_loss: 0.2776 - val_mae: 0.6898 - learning_rate: 0.0010

Epoch 18/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 11s 738ms/step - loss: 0.3234 - mae: 0.6731 - val_loss: 0.2647 - val_mae: 0.6598 - learning_rate: 0.0010

Epoch 19/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 21s 832ms/step - loss: 0.3097 - mae: 0.6674 - val_loss: 0.2917 - val_mae: 0.6847 - learning_rate: 0.0010

Epoch 20/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 21s 844ms/step - loss: 0.2624 - mae: 0.6187 - val_loss: 0.2805 - val_mae: 0.6927 - learning_rate: 0.0010

Epoch 21/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 12s 848ms/step - loss: 0.3064 - mae: 0.6935 - val_loss: 0.2620 - val_mae: 0.6587 - learning_rate: 0.0010

Epoch 22/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 20s 847ms/step - loss: 0.3451 - mae: 0.7215 - val_loss: 0.2845 - val_mae: 0.6903 - learning_rate: 0.0010

Epoch 23/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 12s 877ms/step - loss: 0.2080 - mae: 0.5248 - val_loss: 0.2620 - val_mae: 0.6540 - learning_rate: 0.0010

Epoch 24/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 20s 847ms/step - loss: 0.2885 - mae: 0.6691 - val_loss: 0.2352 - val_mae: 0.6060 - learning_rate: 0.0010

Epoch 25/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 18s 650ms/step - loss: 0.2502 - mae: 0.6027 - val_loss: 0.2234 - val_mae: 0.5727 - learning_rate: 0.0010

Epoch 26/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 12s 849ms/step - loss: 0.2379 - mae: 0.5744 - val_loss: 0.2408 - val_mae: 0.6100 - learning_rate: 0.0010

Epoch 27/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 12s 857ms/step - loss: 0.2444 - mae: 0.5959 - val_loss: 0.2564 - val_mae: 0.6482 - learning_rate: 0.0010

Epoch 28/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 20s 849ms/step - loss: 0.2693 - mae: 0.6468 - val_loss: 0.2302 - val_mae: 0.6084 - learning_rate: 0.0010

Epoch 29/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 18s 649ms/step - loss: 0.2294 - mae: 0.5906 - val_loss: 0.2195 - val_mae: 0.5832 - learning_rate: 5.0000e-04

Epoch 30/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 12s 763ms/step - loss: 0.2140 - mae: 0.5521 - val_loss: 0.2144 - val_mae: 0.5630 - learning_rate: 5.0000e-04

Epoch 31/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 20s 772ms/step - loss: 0.2879 - mae: 0.6550 - val_loss: 0.2140 - val_mae: 0.5604 - learning_rate: 5.0000e-04

Epoch 32/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 21s 848ms/step - loss: 0.1955 - mae: 0.5274 - val_loss: 0.2169 - val_mae: 0.5663 - learning_rate: 5.0000e-04

Epoch 33/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 18s 652ms/step - loss: 0.2071 - mae: 0.5200 - val_loss: 0.2178 - val_mae: 0.5703 - learning_rate: 5.0000e-04

Epoch 34/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 12s 850ms/step - loss: 0.1792 - mae: 0.4974 - val_loss: 0.2089 - val_mae: 0.5516 - learning_rate: 5.0000e-04

Epoch 35/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 18s 642ms/step - loss: 0.2178 - mae: 0.5735 - val_loss: 0.2024 - val_mae: 0.5318 - learning_rate: 5.0000e-04

Epoch 36/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 11s 819ms/step - loss: 0.2237 - mae: 0.5832 - val_loss: 0.2029 - val_mae: 0.5412 - learning_rate: 5.0000e-04

Epoch 37/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 12s 845ms/step - loss: 0.2214 - mae: 0.5517 - val_loss: 0.2025 - val_mae: 0.5359 - learning_rate: 5.0000e-04

Epoch 38/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 20s 846ms/step - loss: 0.1960 - mae: 0.5301 - val_loss: 0.2093 - val_mae: 0.5545 - learning_rate: 5.0000e-04

Epoch 39/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 12s 889ms/step - loss: 0.1941 - mae: 0.5460 - val_loss: 0.2100 - val_mae: 0.5498 - learning_rate: 2.5000e-04

Epoch 40/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 20s 854ms/step - loss: 0.1738 - mae: 0.4803 - val_loss: 0.2053 - val_mae: 0.5411 - learning_rate: 2.5000e-04

Fold 4 results:

Test MSE: 0.4319

Test MAE: 0.8072

Test R^2: -0.0795

Per-bin evaluation:

Bin 0 (n=21): MSE=0.6639, MAE=0.7444

Bin 1 (n=14): MSE=0.1554, MAE=0.3310

Bin 2 (n=3): MSE=1.4882, MAE=1.0944

Bin 3 (n=5): MSE=5.0620, MAE=2.2323

Predictions saved to /content/drive/MyDrive/test_predictions_regression_fold_4.csv

Sample predictions:

True: 0.00, Predicted: 0.70

True: 1.00, Predicted: 1.10

True: 1.00, Predicted: 0.48

True: 0.00, Predicted: 0.77

True: 0.00, Predicted: 0.93

True: 0.00, Predicted: 0.96

True: 0.00, Predicted: 1.01

True: 1.00, Predicted: 0.45

True: 3.00, Predicted: 0.46

True: 0.00, Predicted: 1.44

Processing fold 5/5

Sample weight distribution: (array([0.89218834, 1.44921525]), array([225, 54]))

Training samples: 279

Test samples: 43

Training label distribution (rounded): [75 54 75 75]

Test label distribution (rounded): [19 9 9 6]

Model architecture:

Model: "functional_9"

```
┏━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━┳━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━
━┳━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━┳━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━┓
┃ Layer (type)        ┃ Output Shape        ┃     Param # ┃ Connected to        ┃
┡━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╇━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━
━╇━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╇━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━┩
│ mfcc_input          │ (None, 500, 101)  │           0 │ -                 │
│ (InputLayer)        │                   │             │                   │
├─────────────────────────────────────────┼─────────────────────────────────────┼────────────────────────┼────────
────────────────────────────┤
│ egemaps_input       │ (None, 500, 101)  │           0 │ -                 │
│ (InputLayer)        │                   │             │                   │
├─────────────────────────────────────────┼─────────────────────────────────────┼────────────────────────┼────────
────────────────────────────┤
│ not_equal_12        │ (None, 500, 101)  │           0 │ mfcc_input[0][0]  │
│ (NotEqual)          │                   │             │                   │
├─────────────────────────────────────────┼─────────────────────────────────────┼────────────────────────┼────────
────────────────────────────┤
│ not_equal_13        │ (None, 500, 101)  │           0 │ egemaps_input[0]… │
│ (NotEqual)          │                   │             │                   │
├─────────────────────────────────────────┼─────────────────────────────────────┼────────────────────────┼────────
────────────────────────────┤
│ masking_12          │ (None, 500, 101)  │           0 │ mfcc_input[0][0]  │
│ (Masking)           │                   │             │                   │
├─────────────────────────────────────────┼─────────────────────────────────────┼────────────────────────┼────────
────────────────────────────┤
│ any_12 (Any)        │ (None, 500)       │           0 │ not_equal_12[0][… │
```

```
├──────────────────────┼───────────────────┼────────────┼──────
──────────────┤
│ masking_13          │ (None, 500, 101)  │          0 │ egemaps_input[0]… │
│ (Masking)           │                   │            │                   │
├──────────────────────┼───────────────────┼────────────┼──────
──────────────┤
│ any_13 (Any)        │ (None, 500)       │          0 │ not_equal_13[0][… │
├──────────────────────┼───────────────────┼────────────┼──────
──────────────┤
│ lstm_26 (LSTM)      │ (None, 64)        │     42,496 │ masking_12[0][0], │
│                     │                   │            │ any_12[0][0]      │
├──────────────────────┼───────────────────┼────────────┼──────
──────────────┤
│ lstm_27 (LSTM)      │ (None, 64)        │     42,496 │ masking_13[0][0], │
│                     │                   │            │ any_13[0][0]      │
├──────────────────────┼───────────────────┼────────────┼──────
──────────────┤
│ batch_normalizatio… │ (None, 64)        │        256 │ lstm_26[0][0]     │
│ (BatchNormalizatio… │                   │            │                   │
├──────────────────────┼───────────────────┼────────────┼──────
──────────────┤
│ batch_normalizatio… │ (None, 64)        │        256 │ lstm_27[0][0]     │
│ (BatchNormalizatio… │                   │            │                   │
├──────────────────────┼───────────────────┼────────────┼──────
──────────────┤
│ dropout_20          │ (None, 64)        │          0 │ batch_normalizat… │
│ (Dropout)           │                   │            │                   │
```

```
├──────────────────────────┼────────────────────────┼───────────────┼─────
───────────────────┤
│ dropout_21          │ (None, 64)          │          0 │ batch_normalizat… │
│ (Dropout)           │                     │            │                   │
├──────────────────────────┼────────────────────────┼───────────────┼─────
───────────────────┤
│ concatenate_9       │ (None, 128)         │          0 │ dropout_20[0][0], │
│ (Concatenate)       │                     │            │ dropout_21[0][0]  │
├──────────────────────────┼────────────────────────┼───────────────┼─────
───────────────────┤
│ dense_14 (Dense)    │ (None, 32)          │      4,128 │ concatenate_9[0]… │
├──────────────────────────┼────────────────────────┼───────────────┼─────
───────────────────┤
│ dropout_22          │ (None, 32)          │          0 │ dense_14[0][0]    │
│ (Dropout)           │                     │            │                   │
├──────────────────────────┼────────────────────────┼───────────────┼─────
───────────────────┤
│ dense_15 (Dense)    │ (None, 1)           │         33 │ dropout_22[0][0]  │
└──────────────────────────┴────────────────────────┴───────────────┴─────
───────────────────┘
 Total params: 89,665 (350.25 KB)
 Trainable params: 89,409 (349.25 KB)
 Non-trainable params: 256 (1.00 KB)
Epoch 1/50
14/14 ━━━━━━━━━━━━━━━━━━━━ 22s 917ms/step - loss: 1.8478 - mae: 2.3551 - val_loss: 0.8697 - val_mae: 1.3505 - learning_rate: 0.0010
Epoch 2/50
```

```
14/14 ━━━━━━━━━━━━━━━━━━━━ 19s 795ms/step - loss: 1.1807 - mae: 1.6557 - val_loss: 0.7080 - val_mae: 1.1871 - learning_rate: 0.0010
Epoch 3/50
14/14 ━━━━━━━━━━━━━━━━━━━━ 18s 630ms/step - loss: 0.9776 - mae: 1.4052 - val_loss: 0.5721 - val_mae: 1.0486 - learning_rate: 0.0010
Epoch 4/50
14/14 ━━━━━━━━━━━━━━━━━━━━ 12s 729ms/step - loss: 0.7757 - mae: 1.2051 - val_loss: 0.5040 - val_mae: 0.9714 - learning_rate: 0.0010
Epoch 5/50
14/14 ━━━━━━━━━━━━━━━━━━━━ 21s 775ms/step - loss: 0.8212 - mae: 1.2413 - val_loss: 0.4812 - val_mae: 0.9548 - learning_rate: 0.0010
Epoch 6/50
14/14 ━━━━━━━━━━━━━━━━━━━━ 22s 854ms/step - loss: 0.6523 - mae: 1.0835 - val_loss: 0.4306 - val_mae: 0.8795 - learning_rate: 0.0010
Epoch 7/50
14/14 ━━━━━━━━━━━━━━━━━━━━ 11s 810ms/step - loss: 0.6175 - mae: 1.0038 - val_loss: 0.4148 - val_mae: 0.8483 - learning_rate: 0.0010
Epoch 8/50
14/14 ━━━━━━━━━━━━━━━━━━━━ 21s 839ms/step - loss: 0.4727 - mae: 0.8645 - val_loss: 0.3653 - val_mae: 0.7916 - learning_rate: 0.0010
Epoch 9/50
14/14 ━━━━━━━━━━━━━━━━━━━━ 11s 813ms/step - loss: 0.5886 - mae: 1.0198 - val_loss: 0.3153 - val_mae: 0.7406 - learning_rate: 0.0010
Epoch 10/50
14/14 ━━━━━━━━━━━━━━━━━━━━ 21s 842ms/step - loss: 0.5323 - mae: 0.9171 - val_loss: 0.2933 - val_mae: 0.7031 - learning_rate: 0.0010
Epoch 11/50
```

14/14 ━━━━━━━━━━━━━━━━━━━━ 11s 815ms/step - loss: 0.5711 - mae: 0.9564 - val_loss: 0.3236 - val_mae: 0.7232 - learning_rate: 0.0010

Epoch 12/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 9s 650ms/step - loss: 0.5153 - mae: 0.9080 - val_loss: 0.3253 - val_mae: 0.7058 - learning_rate: 0.0010

Epoch 13/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 13s 827ms/step - loss: 0.4678 - mae: 0.8908 - val_loss: 0.2722 - val_mae: 0.6387 - learning_rate: 0.0010

Epoch 14/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 18s 643ms/step - loss: 0.4950 - mae: 0.8999 - val_loss: 0.2541 - val_mae: 0.6361 - learning_rate: 0.0010

Epoch 15/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 12s 747ms/step - loss: 0.4302 - mae: 0.8204 - val_loss: 0.2641 - val_mae: 0.6625 - learning_rate: 0.0010

Epoch 16/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 12s 843ms/step - loss: 0.3624 - mae: 0.7353 - val_loss: 0.2376 - val_mae: 0.6111 - learning_rate: 0.0010

Epoch 17/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 9s 646ms/step - loss: 0.4683 - mae: 0.8461 - val_loss: 0.2217 - val_mae: 0.5564 - learning_rate: 0.0010

Epoch 18/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 11s 681ms/step - loss: 0.3781 - mae: 0.7497 - val_loss: 0.2091 - val_mae: 0.5332 - learning_rate: 0.0010

Epoch 19/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 12s 839ms/step - loss: 0.4022 - mae: 0.7892 - val_loss: 0.2278 - val_mae: 0.5646 - learning_rate: 0.0010

Epoch 20/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 19s 706ms/step - loss: 0.3648 - mae: 0.7338 - val_loss: 0.2297 - val_mae: 0.5779 - learning_rate: 0.0010

Epoch 21/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 12s 852ms/step - loss: 0.3667 - mae: 0.7454 - val_loss: 0.2201 - val_mae: 0.5708 - learning_rate: 0.0010

Epoch 22/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 10s 734ms/step - loss: 0.3462 - mae: 0.7297 - val_loss: 0.2114 - val_mae: 0.5649 - learning_rate: 5.0000e-04

Epoch 23/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 10s 687ms/step - loss: 0.2845 - mae: 0.6280 - val_loss: 0.2058 - val_mae: 0.5610 - learning_rate: 5.0000e-04

Epoch 24/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 12s 836ms/step - loss: 0.2852 - mae: 0.6330 - val_loss: 0.1955 - val_mae: 0.5391 - learning_rate: 5.0000e-04

Epoch 25/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 19s 700ms/step - loss: 0.3111 - mae: 0.6532 - val_loss: 0.2036 - val_mae: 0.5357 - learning_rate: 5.0000e-04

Epoch 26/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 12s 845ms/step - loss: 0.3371 - mae: 0.6997 - val_loss: 0.1877 - val_mae: 0.5132 - learning_rate: 5.0000e-04

Epoch 27/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 19s 702ms/step - loss: 0.3177 - mae: 0.6626 - val_loss: 0.1879 - val_mae: 0.5132 - learning_rate: 5.0000e-04

Epoch 28/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 21s 781ms/step - loss: 0.2448 - mae: 0.5729 - val_loss: 0.1876 - val_mae: 0.5080 - learning_rate: 5.0000e-04

Epoch 29/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 21s 841ms/step - loss: 0.3127 - mae: 0.6874 - val_loss: 0.1912 - val_mae: 0.5195 - learning_rate: 5.0000e-04

Epoch 30/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 18s 636ms/step - loss: 0.2793 - mae: 0.6290 - val_loss: 0.1845 - val_mae: 0.4970 - learning_rate: 5.0000e-04

Epoch 31/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 13s 790ms/step - loss: 0.2656 - mae: 0.6041 - val_loss: 0.1760 - val_mae: 0.4700 - learning_rate: 5.0000e-04

Epoch 32/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 18s 643ms/step - loss: 0.3194 - mae: 0.6787 - val_loss: 0.1745 - val_mae: 0.4535 - learning_rate: 5.0000e-04

Epoch 33/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 11s 794ms/step - loss: 0.2850 - mae: 0.6356 - val_loss: 0.1707 - val_mae: 0.4515 - learning_rate: 5.0000e-04

Epoch 34/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 18s 654ms/step - loss: 0.2413 - mae: 0.5909 - val_loss: 0.1702 - val_mae: 0.4688 - learning_rate: 5.0000e-04

Epoch 35/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 11s 773ms/step - loss: 0.2649 - mae: 0.6013 - val_loss: 0.1742 - val_mae: 0.4759 - learning_rate: 5.0000e-04

Epoch 36/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 12s 845ms/step - loss: 0.2683 - mae: 0.6238 - val_loss: 0.1727 - val_mae: 0.4540 - learning_rate: 5.0000e-04

Epoch 37/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 20s 766ms/step - loss: 0.2391 - mae: 0.5546 - val_loss: 0.1710 - val_mae: 0.4597 - learning_rate: 5.0000e-04

Epoch 38/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 20s 728ms/step - loss: 0.2627 - mae: 0.6135 - val_loss: 0.1688 - val_mae: 0.4540 - learning_rate: 2.5000e-04

Epoch 39/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 10s 698ms/step - loss: 0.2545 - mae: 0.6044 - val_loss: 0.1666 - val_mae: 0.4418 - learning_rate: 2.5000e-04

Epoch 40/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 12s 846ms/step - loss: 0.2448 - mae: 0.5721 - val_loss: 0.1635 - val_mae: 0.4372 - learning_rate: 2.5000e-04

Epoch 41/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 10s 746ms/step - loss: 0.2293 - mae: 0.5760 - val_loss: 0.1625 - val_mae: 0.4335 - learning_rate: 2.5000e-04

Epoch 42/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 22s 825ms/step - loss: 0.2375 - mae: 0.5575 - val_loss: 0.1656 - val_mae: 0.4400 - learning_rate: 2.5000e-04

Epoch 43/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 18s 643ms/step - loss: 0.2010 - mae: 0.5235 - val_loss: 0.1673 - val_mae: 0.4338 - learning_rate: 2.5000e-04

Epoch 44/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 12s 845ms/step - loss: 0.2026 - mae: 0.5280 - val_loss: 0.1678 - val_mae: 0.4423 - learning_rate: 2.5000e-04

Epoch 45/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 18s 640ms/step - loss: 0.2124 - mae: 0.5438 - val_loss: 0.1683 - val_mae: 0.4471 - learning_rate: 1.2500e-04

Epoch 46/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 12s 843ms/step - loss: 0.2129 - mae: 0.5145 - val_loss: 0.1654 - val_mae: 0.4375 - learning_rate: 1.2500e-04

Fold 5 results:

Test MSE: 0.6295

Test MAE: 1.0614

Test R^2: -0.3249

Per-bin evaluation:

Bin 0 (n=19): MSE=0.9285, MAE=0.8577

Bin 1 (n=9): MSE=0.3795, MAE=0.5268

Bin 2 (n=9): MSE=1.5456, MAE=1.1638

Bin 3 (n=6): MSE=5.6341, MAE=2.3548

Predictions saved to /content/drive/MyDrive/test_predictions_regression_fold_5.csv

Sample predictions:

True: 2.00, Predicted: 1.25

True: 0.00, Predicted: 1.35

True: 0.00, Predicted: 0.61

True: 2.00, Predicted: 0.51

True: 1.00, Predicted: 0.28

True: 3.00, Predicted: 0.47

True: 3.00, Predicted: 1.27

True: 2.00, Predicted: 1.00

True: 3.00, Predicted: 0.73

True: 0.00, Predicted: 0.32

Cross-validation results: MSE=0.5468±0.0870, MAE=0.9492±0.0993, R^2=-0.3624±0.2038

Pipeline execution completed!

Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).

Starting appetite prediction pipeline with cross-validation...

Loading data from Google Drive folders...

Files in /content/drive/MyDrive/audio_files/AA:

['audio_files.html', '713_BoAW_openSMILE_2.3.0_eGeMAPS.csv', '687_BoAW_openSMILE_2.3.0_eGeMAPS.csv', '670_BoAW_openSMILE_2.3.0_eGeMAPS.csv', '698_BoAW_openSMILE_2.3.0_eGeMAPS.csv', '632_BoAW_openSMILE_2.3.0_eGeMAPS.csv', '667_BoAW_openSMILE_2.3.0_eGeMAPS.csv', '653_BoAW_openSMILE_2.3.0_eGeMAPS.csv', '657_BoAW_openSMILE_2.3.0_eGeMAPS.csv', '718_BoAW_openSMILE_2.3.0_eGeMAPS.csv'] ...

Files in /content/drive/MyDrive/merged participants:

['merged data of participants', 'Copy of appetite levels mapping.csv', 'appetite levels mapping.csv']

Using mapping file: /content/drive/MyDrive/merged participants/Copy of appetite levels mapping.csv

Mapping file loaded successfully

First few rows of mapping file:

	Participant	PHQ8_5_Appetite	Appetite_Label
0	300	1	loss_or_over
1	301	1	loss_or_over
2	302	0	normal
3	303	0	normal
4	304	2	loss_or_over

Columns: ['Participant', 'PHQ8_5_Appetite', 'Appetite_Label']

PHQ8_5_Appetite value counts:

PHQ8_5_Appetite

0   115

1   83

2   44

3   33

Name: count, dtype: int64

Loaded labels for 275 participants

Sample participant labels: [('300', 1), ('301', 1), ('302', 0), ('303', 0), ('304', 2)]

Found 433 CSV files in the participant folder

Found 216 MFCC files and 216 eGeMAPS files

Found 216 complete participant records

Processing participants: 31%|███ | 66/216 [00:08<00:06, 23.50it/s]Skipping participant 718 - empty MFCC data

Processing participants: 100%|██████████| 216/216 [00:16<00:00, 12.84it/s]

Data preprocessing complete

MFCC data shape: (215, 500, 101)

eGeMAPS data shape: (215, 500, 101)

Labels shape: (215,)

Label distribution (rounded): [94 63 34 24]

Processing fold 1/5

Sample weight distribution: (array([0.78991145, 0.90151957, 1.63998653]), array([82, 150, 50]))

Training samples: 282

Test samples: 43

Training label distribution (rounded): [82 50 75 75]

Test label distribution (rounded): [12 13 12 6]

Model architecture:

Model: "functional_5"

```
┏━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━┳━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━
━━┳━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━┳━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━┓
┃ Layer (type)        ┃ Output Shape      ┃   Param # ┃ Connected to      ┃
┡━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╇━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━
━╇━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╇━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━┩
│ mfcc_input          │ (None, 500, 101) │         0 │ -                 │
│ (InputLayer)        │                   │           │                   │
├─────────────────────────────────────────────┼───────────────────────────────────────────┼───────────────────────────┼──────────
──────────────────────────────┤
│ egemaps_input       │ (None, 500, 101) │         0 │ -                 │
│ (InputLayer)        │                   │           │                   │
├─────────────────────────────────────────────┼───────────────────────────────────────────┼───────────────────────────┼──────────
──────────────────────────────┤
│ not_equal_4         │ (None, 500, 101) │         0 │ mfcc_input[0][0]  │
│ (NotEqual)          │                   │           │                   │
├─────────────────────────────────────────────┼───────────────────────────────────────────┼───────────────────────────┼──────────
──────────────────────────────┤
│ not_equal_5         │ (None, 500, 101) │         0 │ egemaps_input[0]… │
│ (NotEqual)          │                   │           │                   │
```

```
├──────────────────────┼──────────────────────┼──────────┼──────
──────────────┤
│ masking_4 (Masking) │ (None, 500, 101) │        0 │ mfcc_input[0][0] │
├──────────────────────┼──────────────────────┼──────────┼──────
──────────────┤
│ any_4 (Any)         │ (None, 500)       │        0 │ not_equal_4[0][0] │
├──────────────────────┼──────────────────────┼──────────┼──────
──────────────┤
│ masking_5 (Masking) │ (None, 500, 101) │        0 │ egemaps_input[0]… │
├──────────────────────┼──────────────────────┼──────────┼──────
──────────────┤
│ any_5 (Any)         │ (None, 500)       │        0 │ not_equal_5[0][0] │
├──────────────────────┼──────────────────────┼──────────┼──────
──────────────┤
│ lstm_18 (LSTM)      │ (None, 64)        │   42,496 │ masking_4[0][0], │
│                     │                   │          │ any_4[0][0]       │
├──────────────────────┼──────────────────────┼──────────┼──────
──────────────┤
│ lstm_19 (LSTM)      │ (None, 64)        │   42,496 │ masking_5[0][0], │
│                     │                   │          │ any_5[0][0]       │
├──────────────────────┼──────────────────────┼──────────┼──────
──────────────┤
│ batch_normalizatio… │ (None, 64)        │      256 │ lstm_18[0][0]     │
│ (BatchNormalizatio… │                   │          │                   │
├──────────────────────┼──────────────────────┼──────────┼──────
──────────────┤
│ batch_normalizatio… │ (None, 64)        │      256 │ lstm_19[0][0]     │
```

```
│ (BatchNormalizatio… │                     │            │                   │
├─────────────────────┼──────────────────────┼─────────────┼─────
────────────────────┤
│ dropout_8 (Dropout) │ (None, 64)      │       0 │ batch_normalizat… │
├─────────────────────┼──────────────────────┼─────────────┼─────
────────────────────┤
│ dropout_9 (Dropout) │ (None, 64)      │       0 │ batch_normalizat… │
├─────────────────────┼──────────────────────┼─────────────┼─────
────────────────────┤
│ concatenate_5       │ (None, 128)     │       0 │ dropout_8[0][0],  │
│ (Concatenate)       │                 │         │ dropout_9[0][0]   │
├─────────────────────┼──────────────────────┼─────────────┼─────
────────────────────┤
│ dense_6 (Dense)     │ (None, 32)      │   4,128 │ concatenate_5[0]… │
├─────────────────────┼──────────────────────┼─────────────┼─────
────────────────────┤
│ dropout_10          │ (None, 32)      │       0 │ dense_6[0][0]     │
│ (Dropout)           │                 │         │                   │
├─────────────────────┼──────────────────────┼─────────────┼─────
────────────────────┤
│ dense_7 (Dense)     │ (None, 1)       │      33 │ dropout_10[0][0]  │
└─────────────────────┴──────────────────────┴─────────────┴─────
────────────────────┘
 Total params: 89,665 (350.25 KB)
 Trainable params: 89,409 (349.25 KB)
 Non-trainable params: 256 (1.00 KB)
Epoch 1/50
```

```
15/15 ━━━━━━━━━━━━━━━━━━━━ 23s 835ms/step - loss: 1.4101 - mae: 1.8750 - val_loss: 0.9297 - val_mae: 1.3902 - learning_rate: 0.0010
Epoch 2/50
15/15 ━━━━━━━━━━━━━━━━━━━━ 20s 820ms/step - loss: 1.1258 - mae: 1.5644 - val_loss: 0.6970 - val_mae: 1.1612 - learning_rate: 0.0010
Epoch 3/50
15/15 ━━━━━━━━━━━━━━━━━━━━ 19s 677ms/step - loss: 0.8261 - mae: 1.2990 - val_loss: 0.5943 - val_mae: 1.0614 - learning_rate: 0.0010
Epoch 4/50
15/15 ━━━━━━━━━━━━━━━━━━━━ 12s 824ms/step - loss: 0.6006 - mae: 1.0333 - val_loss: 0.5763 - val_mae: 1.0428 - learning_rate: 0.0010
Epoch 5/50
15/15 ━━━━━━━━━━━━━━━━━━━━ 18s 646ms/step - loss: 0.7918 - mae: 1.2332 - val_loss: 0.4935 - val_mae: 0.9484 - learning_rate: 0.0010
Epoch 6/50
15/15 ━━━━━━━━━━━━━━━━━━━━ 12s 807ms/step - loss: 0.6948 - mae: 1.1211 - val_loss: 0.4634 - val_mae: 0.9127 - learning_rate: 0.0010
Epoch 7/50
15/15 ━━━━━━━━━━━━━━━━━━━━ 18s 638ms/step - loss: 0.5265 - mae: 0.9470 - val_loss: 0.4555 - val_mae: 0.9061 - learning_rate: 0.0010
Epoch 8/50
15/15 ━━━━━━━━━━━━━━━━━━━━ 12s 807ms/step - loss: 0.5533 - mae: 0.9547 - val_loss: 0.4244 - val_mae: 0.8684 - learning_rate: 0.0010
Epoch 9/50
15/15 ━━━━━━━━━━━━━━━━━━━━ 18s 629ms/step - loss: 0.5331 - mae: 0.9489 - val_loss: 0.3232 - val_mae: 0.7527 - learning_rate: 0.0010
Epoch 10/50
```

15/15 ━━━━━━━━━━━━━━━━━━━━ 12s 732ms/step - loss: 0.4084 - mae: 0.7768 - val_loss: 0.2960 - val_mae: 0.7134 - learning_rate: 0.0010

Epoch 11/50

15/15 ━━━━━━━━━━━━━━━━━━━━ 20s 760ms/step - loss: 0.4385 - mae: 0.8543 - val_loss: 0.2781 - val_mae: 0.6900 - learning_rate: 0.0010

Epoch 12/50

15/15 ━━━━━━━━━━━━━━━━━━━━ 21s 820ms/step - loss: 0.3812 - mae: 0.7651 - val_loss: 0.2891 - val_mae: 0.6893 - learning_rate: 0.0010

Epoch 13/50

15/15 ━━━━━━━━━━━━━━━━━━━━ 18s 625ms/step - loss: 0.4228 - mae: 0.8142 - val_loss: 0.2642 - val_mae: 0.6553 - learning_rate: 0.0010

Epoch 14/50

15/15 ━━━━━━━━━━━━━━━━━━━━ 12s 809ms/step - loss: 0.4020 - mae: 0.7612 - val_loss: 0.2173 - val_mae: 0.6015 - learning_rate: 0.0010

Epoch 15/50

15/15 ━━━━━━━━━━━━━━━━━━━━ 18s 617ms/step - loss: 0.4308 - mae: 0.8365 - val_loss: 0.2726 - val_mae: 0.6492 - learning_rate: 0.0010

Epoch 16/50

15/15 ━━━━━━━━━━━━━━━━━━━━ 12s 807ms/step - loss: 0.3869 - mae: 0.7807 - val_loss: 0.2440 - val_mae: 0.6143 - learning_rate: 0.0010

Epoch 17/50

15/15 ━━━━━━━━━━━━━━━━━━━━ 18s 627ms/step - loss: 0.3877 - mae: 0.7872 - val_loss: 0.2338 - val_mae: 0.5954 - learning_rate: 0.0010

Epoch 18/50

15/15 ━━━━━━━━━━━━━━━━━━━━ 12s 726ms/step - loss: 0.3393 - mae: 0.7145 - val_loss: 0.2138 - val_mae: 0.5677 - learning_rate: 5.0000e-04

Epoch 19/50

15/15 ━━━━━━━━━━━━━━━━━━━━ 20s 764ms/step - loss: 0.3407 - mae: 0.7144 - val_loss: 0.2180 - val_mae: 0.5619 - learning_rate: 5.0000e-04

Epoch 20/50

15/15 ━━━━━━━━━━━━━━━━━━━━ 21s 817ms/step - loss: 0.3384 - mae: 0.6887 - val_loss: 0.2171 - val_mae: 0.5546 - learning_rate: 5.0000e-04

Epoch 21/50

15/15 ━━━━━━━━━━━━━━━━━━━━ 12s 799ms/step - loss: 0.3671 - mae: 0.7471 - val_loss: 0.2151 - val_mae: 0.5355 - learning_rate: 5.0000e-04

Epoch 22/50

15/15 ━━━━━━━━━━━━━━━━━━━━ 9s 616ms/step - loss: 0.3578 - mae: 0.7045 - val_loss: 0.2158 - val_mae: 0.5313 - learning_rate: 2.5000e-04

Epoch 23/50

15/15 ━━━━━━━━━━━━━━━━━━━━ 13s 755ms/step - loss: 0.2586 - mae: 0.6219 - val_loss: 0.2139 - val_mae: 0.5286 - learning_rate: 2.5000e-04

WARNING:tensorflow:5 out of the last 5 calls to <function TensorFlowTrainer.make_predict_function.<locals>.one_step_on_data_distributed at 0x7f5cacbde5c0> triggered tf.function retracing. Tracing is expensive and the excessive number of tracings could be due to (1) creating @tf.function repeatedly in a loop, (2) passing tensors with different shapes, (3) passing Python objects instead of tensors. For (1), please define your @tf.function outside of the loop. For (2), @tf.function has reduce_retracing=True option that can avoid unnecessary retracing. For (3), please refer to https://www.tensorflow.org/guide/function#controlling_retracing and https://www.tensorflow.org/api_docs/python/tf/function for more details.

WARNING:tensorflow:6 out of the last 6 calls to <function TensorFlowTrainer.make_predict_function.<locals>.one_step_on_data_distributed at 0x7f5cacbde5c0> triggered tf.function retracing. Tracing is expensive and the excessive number of tracings could be due to (1) creating @tf.function repeatedly in a loop, (2) passing tensors with different shapes, (3) passing Python objects instead of tensors. For (1), please define your @tf.function outside of the loop. For (2), @tf.function has reduce_retracing=True option that can avoid unnecessary retracing. For (3), please refer to https://www.tensorflow.org/guide/function#controlling_retracing and https://www.tensorflow.org/api_docs/python/tf/function for more details.

Fold 1 results:

Test MSE: 0.6518

Test MAE: 1.0445

Test R^2: -0.6099

Per-bin evaluation:

Bin 0 (n=12): MSE=0.6553, MAE=0.7572

Bin 1 (n=13): MSE=0.1506, MAE=0.2827

Bin 2 (n=12): MSE=2.3594, MAE=1.5113

Bin 3 (n=6): MSE=5.6252, MAE=2.3358

Predictions saved to /content/drive/MyDrive/test_predictions_regression_fold_1.csv

Sample predictions:

True: 1.00, Predicted: 1.08

True: 2.00, Predicted: 0.36

True: 0.00, Predicted: 0.61

True: 2.00, Predicted: 0.74

True: 3.00, Predicted: 0.64

True: 2.00, Predicted: 0.62

True: 0.00, Predicted: 0.94

True: 0.00, Predicted: 1.20

True: 0.00, Predicted: 1.11

True: 1.00, Predicted: 1.24

Processing fold 2/5

Sample weight distribution: (array([0.85925055, 1.65976304]), array([225, 48]))

Training samples: 273

Test samples: 43

Training label distribution (rounded): [75 48 75 75]

Test label distribution (rounded): [19 15 5 4]

Model architecture:

Model: "functional_6"

```
┏━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━┳━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━
━━┳━━━━━━━━━━━━━━━━━━━━━━━━━┳━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━┓
┃ Layer (type)        ┃ Output Shape      ┃   Param # ┃ Connected to      ┃
┡━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╇━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━
╇━━━━━━━━━━━━━━━━━━━━━━━━━╇━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━┩
│ mfcc_input        │ (None, 500, 101) │         0 │ -             │
│ (InputLayer)      │                  │           │               │
├──────────────────────┼──────────────────────┼──────────────┼──────
───────────────┤
│ egemaps_input     │ (None, 500, 101) │         0 │ -             │
│ (InputLayer)      │                  │           │               │
├──────────────────────┼──────────────────────┼──────────────┼──────
───────────────┤
│ not_equal_6       │ (None, 500, 101) │         0 │ mfcc_input[0][0] │
│ (NotEqual)        │                  │           │               │
```

```
├──────────────────────────┼──────────────────────────┼────────────────┼─────
──────────────────┤
│ not_equal_7         │ (None, 500, 101)  │         0 │ egemaps_input[0]… │
│ (NotEqual)        │                   │           │                   │
├──────────────────────────┼──────────────────────────┼────────────────┼─────
──────────────────┤
│ masking_6 (Masking) │ (None, 500, 101)  │         0 │ mfcc_input[0][0]  │
├──────────────────────────┼──────────────────────────┼────────────────┼─────
──────────────────┤
│ any_6 (Any)         │ (None, 500)       │         0 │ not_equal_6[0][0] │
├──────────────────────────┼──────────────────────────┼────────────────┼─────
──────────────────┤
│ masking_7 (Masking) │ (None, 500, 101)  │         0 │ egemaps_input[0]… │
├──────────────────────────┼──────────────────────────┼────────────────┼─────
──────────────────┤
│ any_7 (Any)         │ (None, 500)       │         0 │ not_equal_7[0][0] │
├──────────────────────────┼──────────────────────────┼────────────────┼─────
──────────────────┤
│ lstm_20 (LSTM)      │ (None, 64)        │    42,496 │ masking_6[0][0],  │
│                     │                   │           │ any_6[0][0]       │
├──────────────────────────┼──────────────────────────┼────────────────┼─────
──────────────────┤
│ lstm_21 (LSTM)      │ (None, 64)        │    42,496 │ masking_7[0][0],  │
│                     │                   │           │ any_7[0][0]       │
├──────────────────────────┼──────────────────────────┼────────────────┼─────
──────────────────┤
│ batch_normalizatio… │ (None, 64)        │       256 │ lstm_20[0][0]     │
```

(BatchNormalizatio…			
batch_normalizatio… (BatchNormalizatio…	(None, 64)	256	lstm_21[0][0]
dropout_11 (Dropout)	(None, 64)	0	batch_normalizat…
dropout_12 (Dropout)	(None, 64)	0	batch_normalizat…
concatenate_6 (Concatenate)	(None, 128)	0	dropout_11[0][0], dropout_12[0][0]
dense_8 (Dense)	(None, 32)	4,128	concatenate_6[0]…
dropout_13 (Dropout)	(None, 32)	0	dense_8[0][0]
dense_9 (Dense)	(None, 1)	33	dropout_13[0][0]

└──────────────────────────┴────────────────────────┴────────────────┴──────

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Total params: 89,665 (350.25 KB)

Trainable params: 89,409 (349.25 KB)

Non-trainable params: 256 (1.00 KB)

Epoch 1/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 22s 884ms/step - loss: 2.6882 - mae: 3.3049 - val_loss: 1.1459 - val_mae: 1.6673 - learning_rate: 0.0010

Epoch 2/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 9s 620ms/step - loss: 1.9328 - mae: 2.4670 - val_loss: 0.9549 - val_mae: 1.4466 - learning_rate: 0.0010

Epoch 3/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 12s 685ms/step - loss: 1.3292 - mae: 1.8788 - val_loss: 0.7216 - val_mae: 1.2179 - learning_rate: 0.0010

Epoch 4/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 11s 810ms/step - loss: 1.2025 - mae: 1.6564 - val_loss: 0.6115 - val_mae: 1.1057 - learning_rate: 0.0010

Epoch 5/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 9s 663ms/step - loss: 0.9170 - mae: 1.3404 - val_loss: 0.5708 - val_mae: 1.0584 - learning_rate: 0.0010

Epoch 6/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 10s 629ms/step - loss: 0.9217 - mae: 1.3813 - val_loss: 0.5461 - val_mae: 1.0310 - learning_rate: 0.0010

Epoch 7/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 11s 814ms/step - loss: 0.8368 - mae: 1.2425 - val_loss: 0.4937 - val_mae: 0.9765 - learning_rate: 0.0010

Epoch 8/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 19s 661ms/step - loss: 0.5882 - mae: 1.0112 - val_loss: 0.4640 - val_mae: 0.9378 - learning_rate: 0.0010

Epoch 9/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 12s 819ms/step - loss: 0.6384 - mae: 1.0289 - val_loss: 0.4513 - val_mae: 0.9054 - learning_rate: 0.0010

Epoch 10/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 19s 662ms/step - loss: 0.7428 - mae: 1.1777 - val_loss: 0.4087 - val_mae: 0.8543 - learning_rate: 0.0010

Epoch 11/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 11s 825ms/step - loss: 0.6994 - mae: 1.1114 - val_loss: 0.4007 - val_mae: 0.8446 - learning_rate: 0.0010

Epoch 12/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 19s 678ms/step - loss: 0.6186 - mae: 1.0169 - val_loss: 0.3963 - val_mae: 0.8460 - learning_rate: 0.0010

Epoch 13/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 12s 818ms/step - loss: 0.5848 - mae: 0.9993 - val_loss: 0.3893 - val_mae: 0.8152 - learning_rate: 0.0010

Epoch 14/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 19s 696ms/step - loss: 0.6229 - mae: 1.0112 - val_loss: 0.3741 - val_mae: 0.7922 - learning_rate: 0.0010

Epoch 15/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 11s 820ms/step - loss: 0.5016 - mae: 0.9012 - val_loss: 0.3593 - val_mae: 0.7861 - learning_rate: 0.0010

Epoch 16/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 9s 633ms/step - loss: 0.5608 - mae: 1.0097 - val_loss: 0.3554 - val_mae: 0.7630 - learning_rate: 0.0010

Epoch 17/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 11s 675ms/step - loss: 0.4986 - mae: 0.8924 - val_loss: 0.3307 - val_mae: 0.7163 - learning_rate: 0.0010

Epoch 18/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 11s 807ms/step - loss: 0.4684 - mae: 0.8735 - val_loss: 0.3032 - val_mae: 0.6820 - learning_rate: 0.0010

Epoch 19/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 9s 676ms/step - loss: 0.5117 - mae: 0.9416 - val_loss: 0.3028 - val_mae: 0.6856 - learning_rate: 0.0010

Epoch 20/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 11s 685ms/step - loss: 0.4888 - mae: 0.8746 - val_loss: 0.3009 - val_mae: 0.6648 - learning_rate: 0.0010

Epoch 21/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 11s 814ms/step - loss: 0.3800 - mae: 0.7670 - val_loss: 0.2683 - val_mae: 0.6356 - learning_rate: 0.0010

Epoch 22/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 19s 687ms/step - loss: 0.3159 - mae: 0.6836 - val_loss: 0.2916 - val_mae: 0.6729 - learning_rate: 0.0010

Epoch 23/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 11s 755ms/step - loss: 0.3398 - mae: 0.7146 - val_loss: 0.2695 - val_mae: 0.6635 - learning_rate: 0.0010

Epoch 24/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 19s 620ms/step - loss: 0.3738 - mae: 0.7546 - val_loss: 0.2748 - val_mae: 0.6531 - learning_rate: 0.0010

Epoch 25/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 11s 782ms/step - loss: 0.3225 - mae: 0.6812 - val_loss: 0.2781 - val_mae: 0.6378 - learning_rate: 5.0000e-04

Epoch 26/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 20s 715ms/step - loss: 0.3005 - mae: 0.6726 - val_loss: 0.2694 - val_mae: 0.6179 - learning_rate: 5.0000e-04

Fold 2 results:

Test MSE: 0.4649

Test MAE: 0.8647

Test R^2: -0.2223

Per-bin evaluation:

Bin 0 (n=19): MSE=0.8662, MAE=0.8471

Bin 1 (n=15): MSE=0.3069, MAE=0.4492

Bin 2 (n=5): MSE=1.5566, MAE=1.2272

Bin 3 (n=4): MSE=4.7562, MAE=2.0540

Predictions saved to /content/drive/MyDrive/test_predictions_regression_fold_2.csv

Sample predictions:

True: 0.00, Predicted: 0.79

True: 3.00, Predicted: 1.69

True: 2.00, Predicted: 0.45

True: 0.00, Predicted: 0.68

True: 1.00, Predicted: 0.46

True: 3.00, Predicted: -0.20

True: 1.00, Predicted: 0.41

True: 1.00, Predicted: 0.61

True: 1.00, Predicted: 0.55

True: 3.00, Predicted: 0.84

Processing fold 3/5

Sample weight distribution: (array([0.85774462, 0.93020709, 1.51556084]), array([150, 71, 51]))

Training samples: 272

Test samples: 43

Training label distribution (rounded): [71 51 75 75]

Test label distribution (rounded): [23 12 5 3]

Model architecture:

Model: "functional_7"

```
┏━━━━━━━━━━━━━━━━━━━━━┳━━━━━━━━━━━━━━━━━━━┳━━━━━━━━━━━━┳━━━━━━━━━━━━━━━━━━━┓
┃ Layer (type)        ┃ Output Shape      ┃    Param # ┃ Connected to      ┃
┡━━━━━━━━━━━━━━━━━━━━━╇━━━━━━━━━━━━━━━━━━━╇━━━━━━━━━━━━╇━━━━━━━━━━━━━━━━━━━┩
│ mfcc_input          │ (None, 500, 101)  │          0 │ -                 │
│ (InputLayer)        │                   │            │                   │
├─────────────────────┼───────────────────┼────────────┼───────────────────┤
│ egemaps_input       │ (None, 500, 101)  │          0 │ -                 │
│ (InputLayer)        │                   │            │                   │
├─────────────────────┼───────────────────┼────────────┼───────────────────┤
│ not_equal_8         │ (None, 500, 101)  │          0 │ mfcc_input[0][0]  │
```

(NotEqual)			
not_equal_9	(None, 500, 101)	0	egemaps_input[0]…
(NotEqual)			
masking_8 (Masking)	(None, 500, 101)	0	mfcc_input[0][0]
any_8 (Any)	(None, 500)	0	not_equal_8[0][0]
masking_9 (Masking)	(None, 500, 101)	0	egemaps_input[0]…
any_9 (Any)	(None, 500)	0	not_equal_9[0][0]
lstm_22 (LSTM)	(None, 64)	42,496	masking_8[0][0],
			any_8[0][0]
lstm_23 (LSTM)	(None, 64)	42,496	masking_9[0][0],
			any_9[0][0]

```
│ batch_normalizatio… │ (None, 64)        │        256 │ lstm_22[0][0]     │
│ (BatchNormalizatio… │                   │            │                   │
├─────────────────────┼───────────────────┼────────────┼───────────────────┤
│ batch_normalizatio… │ (None, 64)        │        256 │ lstm_23[0][0]     │
│ (BatchNormalizatio… │                   │            │                   │
├─────────────────────┼───────────────────┼────────────┼───────────────────┤
│ dropout_14          │ (None, 64)        │          0 │ batch_normalizat… │
│ (Dropout)           │                   │            │                   │
├─────────────────────┼───────────────────┼────────────┼───────────────────┤
│ dropout_15          │ (None, 64)        │          0 │ batch_normalizat… │
│ (Dropout)           │                   │            │                   │
├─────────────────────┼───────────────────┼────────────┼───────────────────┤
│ concatenate_7       │ (None, 128)       │          0 │ dropout_14[0][0], │
│ (Concatenate)       │                   │            │ dropout_15[0][0]  │
├─────────────────────┼───────────────────┼────────────┼───────────────────┤
│ dense_10 (Dense)    │ (None, 32)        │      4,128 │ concatenate_7[0]… │
├─────────────────────┼───────────────────┼────────────┼───────────────────┤
│ dropout_16          │ (None, 32)        │          0 │ dense_10[0][0]    │
│ (Dropout)           │                   │            │                   │
├─────────────────────┼───────────────────┼────────────┼───────────────────┤
```

```
│ dense_11 (Dense)    │ (None, 1)        │        33 │ dropout_16[0][0]  │
└──────────────────────────┴────────────────────────┴────────────────┴────────
────────────────┘
```

Total params: 89,665 (350.25 KB)

Trainable params: 89,409 (349.25 KB)

Non-trainable params: 256 (1.00 KB)

Epoch 1/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 20s 862ms/step - loss: 1.0785 - mae: 1.6024 - val_loss: 0.6533 - val_mae: 1.1494 - learning_rate: 0.0010

Epoch 2/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 20s 777ms/step - loss: 0.8202 - mae: 1.2626 - val_loss: 0.5875 - val_mae: 1.0717 - learning_rate: 0.0010

Epoch 3/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 11s 786ms/step - loss: 0.5886 - mae: 0.9911 - val_loss: 0.5296 - val_mae: 0.9956 - learning_rate: 0.0010

Epoch 4/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 20s 743ms/step - loss: 0.6128 - mae: 1.0556 - val_loss: 0.4550 - val_mae: 0.9146 - learning_rate: 0.0010

Epoch 5/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 18s 585ms/step - loss: 0.6107 - mae: 1.0444 - val_loss: 0.4326 - val_mae: 0.8854 - learning_rate: 0.0010

Epoch 6/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 11s 616ms/step - loss: 0.5817 - mae: 1.0204 - val_loss: 0.3915 - val_mae: 0.8330 - learning_rate: 0.0010

Epoch 7/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 11s 766ms/step - loss: 0.6301 - mae: 1.0633 - val_loss: 0.3945 - val_mae: 0.8327 - learning_rate: 0.0010

Epoch 8/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 19s 616ms/step - loss: 0.4900 - mae: 0.8896 - val_loss: 0.3848 - val_mae: 0.8366 - learning_rate: 0.0010

Epoch 9/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 12s 774ms/step - loss: 0.4116 - mae: 0.8334 - val_loss: 0.3297 - val_mae: 0.7680 - learning_rate: 0.0010

Epoch 10/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 9s 662ms/step - loss: 0.5134 - mae: 0.9194 - val_loss: 0.3264 - val_mae: 0.7617 - learning_rate: 0.0010

Epoch 11/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 10s 623ms/step - loss: 0.3948 - mae: 0.7543 - val_loss: 0.3152 - val_mae: 0.7455 - learning_rate: 0.0010

Epoch 12/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 11s 822ms/step - loss: 0.3393 - mae: 0.7311 - val_loss: 0.3248 - val_mae: 0.7611 - learning_rate: 0.0010

Epoch 13/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 10s 721ms/step - loss: 0.4370 - mae: 0.8069 - val_loss: 0.3288 - val_mae: 0.7596 - learning_rate: 0.0010

Epoch 14/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 10s 676ms/step - loss: 0.4619 - mae: 0.8281 - val_loss: 0.2622 - val_mae: 0.6590 - learning_rate: 0.0010

Epoch 15/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 12s 836ms/step - loss: 0.3469 - mae: 0.7205 - val_loss: 0.2289 - val_mae: 0.6148 - learning_rate: 0.0010

Epoch 16/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 19s 659ms/step - loss: 0.3493 - mae: 0.7301 - val_loss: 0.2215 - val_mae: 0.6002 - learning_rate: 0.0010

Epoch 17/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 12s 831ms/step - loss: 0.3433 - mae: 0.7250 - val_loss: 0.1945 - val_mae: 0.5495 - learning_rate: 0.0010

Epoch 18/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 13s 978ms/step - loss: 0.2612 - mae: 0.6111 - val_loss: 0.2021 - val_mae: 0.5515 - learning_rate: 0.0010

Epoch 19/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 13s 857ms/step - loss: 0.2916 - mae: 0.6514 - val_loss: 0.2017 - val_mae: 0.5656 - learning_rate: 0.0010

Epoch 20/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 12s 901ms/step - loss: 0.3345 - mae: 0.7181 - val_loss: 0.1877 - val_mae: 0.5322 - learning_rate: 0.0010

Epoch 21/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 10s 746ms/step - loss: 0.2417 - mae: 0.5649 - val_loss: 0.1973 - val_mae: 0.5469 - learning_rate: 0.0010

Epoch 22/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 21s 824ms/step - loss: 0.2979 - mae: 0.6383 - val_loss: 0.1908 - val_mae: 0.5507 - learning_rate: 0.0010

Epoch 23/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 18s 639ms/step - loss: 0.2513 - mae: 0.6086 - val_loss: 0.1755 - val_mae: 0.4949 - learning_rate: 0.0010

Epoch 24/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 11s 824ms/step - loss: 0.2396 - mae: 0.5885 - val_loss: 0.1758 - val_mae: 0.5011 - learning_rate: 0.0010

Epoch 25/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 18s 626ms/step - loss: 0.2669 - mae: 0.6175 - val_loss: 0.1562 - val_mae: 0.4635 - learning_rate: 0.0010

Epoch 26/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 13s 825ms/step - loss: 0.2134 - mae: 0.5593 - val_loss: 0.1507 - val_mae: 0.4413 - learning_rate: 0.0010

Epoch 27/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 12s 875ms/step - loss: 0.1961 - mae: 0.5225 - val_loss: 0.1655 - val_mae: 0.4804 - learning_rate: 0.0010

Epoch 28/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 20s 830ms/step - loss: 0.2331 - mae: 0.5719 - val_loss: 0.1764 - val_mae: 0.4998 - learning_rate: 0.0010

Epoch 29/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 11s 826ms/step - loss: 0.2014 - mae: 0.5342 - val_loss: 0.1637 - val_mae: 0.4597 - learning_rate: 0.0010

Epoch 30/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 9s 635ms/step - loss: 0.2402 - mae: 0.5898 - val_loss: 0.1644 - val_mae: 0.4708 - learning_rate: 5.0000e-04

Epoch 31/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 13s 787ms/step - loss: 0.2225 - mae: 0.5382 - val_loss: 0.1594 - val_mae: 0.4591 - learning_rate: 5.0000e-04

Fold 3 results:

Test MSE: 0.5561

Test MAE: 0.9681

Test R^2: -0.5753

Per-bin evaluation:

Bin 0 (n=23): MSE=1.2001, MAE=0.9962

Bin 1 (n=12): MSE=0.2239, MAE=0.3943

Bin 2 (n=5): MSE=1.9816, MAE=1.3681

Bin 3 (n=3): MSE=5.8462, MAE=2.3813

Predictions saved to /content/drive/MyDrive/test_predictions_regression_fold_3.csv

Sample predictions:

True: 0.00, Predicted: 1.03

True: 1.00, Predicted: 0.69

True: 0.00, Predicted: 0.94

True: 1.00, Predicted: 0.14

True: 0.00, Predicted: 0.71

True: 0.00, Predicted: 1.32

True: 1.00, Predicted: 0.84

True: 1.00, Predicted: 0.55

True: 2.00, Predicted: 0.53

True: 0.00, Predicted: 0.04

Processing fold 4/5

Sample weight distribution: (array([0.85587215, 0.89080306, 1.60388887]), array([150, 73, 49]))

Training samples: 272

Test samples: 43

Training label distribution (rounded): [73 49 75 75]

Test label distribution (rounded): [21 14 3 5]

Model architecture:

Model: "functional_8"

Layer (type)	Output Shape	Param #	Connected to
mfcc_input (InputLayer)	(None, 500, 101)	0	-
egemaps_input (InputLayer)	(None, 500, 101)	0	-
not_equal_10 (NotEqual)	(None, 500, 101)	0	mfcc_input[0][0]
not_equal_11 (NotEqual)	(None, 500, 101)	0	egemaps_input[0]...
masking_10 (Masking)	(None, 500, 101)	0	mfcc_input[0][0]

any_10 (Any)	(None, 500)	0	not_equal_10[0][…
masking_11 (Masking)	(None, 500, 101)	0	egemaps_input[0]…
any_11 (Any)	(None, 500)	0	not_equal_11[0][…
lstm_24 (LSTM)	(None, 64)	42,496	masking_10[0][0], any_10[0][0]
lstm_25 (LSTM)	(None, 64)	42,496	masking_11[0][0], any_11[0][0]
batch_normalizatio… (BatchNormalizatio…	(None, 64)	256	lstm_24[0][0]
batch_normalizatio… (BatchNormalizatio…	(None, 64)	256	lstm_25[0][0]
dropout_17	(None, 64)	0	batch_normalizat…

```
│ (Dropout)         │                │          │               │
├───────────────────┼───────────────────┼─────────────┼────
──────────────┤
│ dropout_18        │ (None, 64)        │         0 │ batch_normalizat… │
│ (Dropout)         │                │          │               │
├───────────────────┼───────────────────┼─────────────┼────
──────────────┤
│ concatenate_8     │ (None, 128)       │         0 │ dropout_17[0][0], │
│ (Concatenate)     │                │          │ dropout_18[0][0]  │
├───────────────────┼───────────────────┼─────────────┼────
──────────────┤
│ dense_12 (Dense)  │ (None, 32)        │     4,128 │ concatenate_8[0]… │
├───────────────────┼───────────────────┼─────────────┼────
──────────────┤
│ dropout_19        │ (None, 32)        │         0 │ dense_12[0][0]    │
│ (Dropout)         │                │          │               │
├───────────────────┼───────────────────┼─────────────┼────
──────────────┤
│ dense_13 (Dense)  │ (None, 1)         │        33 │ dropout_19[0][0]  │
└───────────────────┴───────────────────┴─────────────┴────
──────────────┘
 Total params: 89,665 (350.25 KB)
 Trainable params: 89,409 (349.25 KB)
 Non-trainable params: 256 (1.00 KB)
Epoch 1/50
14/14 ━━━━━━━━━━━━━━━━━━━━ 23s 916ms/step - loss: 1.3874 - mae:
1.9094 - val_loss: 0.7115 - val_mae: 1.2185 - learning_rate: 0.0010
```

Epoch 2/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 17s 641ms/step - loss: 0.9247 - mae: 1.3476 - val_loss: 0.6901 - val_mae: 1.1823 - learning_rate: 0.0010

Epoch 3/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 12s 858ms/step - loss: 0.6785 - mae: 1.1095 - val_loss: 0.6615 - val_mae: 1.1418 - learning_rate: 0.0010

Epoch 4/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 11s 835ms/step - loss: 0.5809 - mae: 0.9880 - val_loss: 0.6360 - val_mae: 1.1046 - learning_rate: 0.0010

Epoch 5/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 21s 849ms/step - loss: 0.5942 - mae: 0.9609 - val_loss: 0.5622 - val_mae: 1.0266 - learning_rate: 0.0010

Epoch 6/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 18s 637ms/step - loss: 0.6413 - mae: 1.0715 - val_loss: 0.5284 - val_mae: 0.9881 - learning_rate: 0.0010

Epoch 7/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 12s 857ms/step - loss: 0.5462 - mae: 0.9934 - val_loss: 0.5541 - val_mae: 1.0079 - learning_rate: 0.0010

Epoch 8/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 18s 635ms/step - loss: 0.4355 - mae: 0.8144 - val_loss: 0.5584 - val_mae: 1.0281 - learning_rate: 0.0010

Epoch 9/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 13s 770ms/step - loss: 0.4355 - mae: 0.8482 - val_loss: 0.5023 - val_mae: 0.9728 - learning_rate: 0.0010

Epoch 10/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 20s 756ms/step - loss: 0.4398 - mae: 0.8279 - val_loss: 0.4808 - val_mae: 0.9462 - learning_rate: 0.0010

Epoch 11/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 22s 849ms/step - loss: 0.3799 - mae: 0.7412 - val_loss: 0.4867 - val_mae: 0.9426 - learning_rate: 0.0010

Epoch 12/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 18s 646ms/step - loss: 0.3874 - mae: 0.7998 - val_loss: 0.4746 - val_mae: 0.9243 - learning_rate: 0.0010

Epoch 13/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 13s 828ms/step - loss: 0.3890 - mae: 0.7767 - val_loss: 0.3866 - val_mae: 0.8240 - learning_rate: 0.0010

Epoch 14/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 12s 852ms/step - loss: 0.3959 - mae: 0.7422 - val_loss: 0.3794 - val_mae: 0.8235 - learning_rate: 0.0010

Epoch 15/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 21s 830ms/step - loss: 0.3815 - mae: 0.7611 - val_loss: 0.4148 - val_mae: 0.8718 - learning_rate: 0.0010

Epoch 16/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 12s 858ms/step - loss: 0.4165 - mae: 0.8301 - val_loss: 0.3467 - val_mae: 0.7847 - learning_rate: 0.0010

Epoch 17/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 9s 675ms/step - loss: 0.3132 - mae: 0.6819 - val_loss: 0.2776 - val_mae: 0.6898 - learning_rate: 0.0010

Epoch 18/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 11s 738ms/step - loss: 0.3234 - mae: 0.6731 - val_loss: 0.2647 - val_mae: 0.6598 - learning_rate: 0.0010

Epoch 19/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 21s 832ms/step - loss: 0.3097 - mae: 0.6674 - val_loss: 0.2917 - val_mae: 0.6847 - learning_rate: 0.0010

Epoch 20/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 21s 844ms/step - loss: 0.2624 - mae: 0.6187 - val_loss: 0.2805 - val_mae: 0.6927 - learning_rate: 0.0010

Epoch 21/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 12s 848ms/step - loss: 0.3064 - mae: 0.6935 - val_loss: 0.2620 - val_mae: 0.6587 - learning_rate: 0.0010

Epoch 22/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 20s 847ms/step - loss: 0.3451 - mae: 0.7215 - val_loss: 0.2845 - val_mae: 0.6903 - learning_rate: 0.0010

Epoch 23/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 12s 877ms/step - loss: 0.2080 - mae: 0.5248 - val_loss: 0.2620 - val_mae: 0.6540 - learning_rate: 0.0010

Epoch 24/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 20s 847ms/step - loss: 0.2885 - mae: 0.6691 - val_loss: 0.2352 - val_mae: 0.6060 - learning_rate: 0.0010

Epoch 25/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 18s 650ms/step - loss: 0.2502 - mae: 0.6027 - val_loss: 0.2234 - val_mae: 0.5727 - learning_rate: 0.0010

Epoch 26/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 12s 849ms/step - loss: 0.2379 - mae: 0.5744 - val_loss: 0.2408 - val_mae: 0.6100 - learning_rate: 0.0010

Epoch 27/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 12s 857ms/step - loss: 0.2444 - mae: 0.5959 - val_loss: 0.2564 - val_mae: 0.6482 - learning_rate: 0.0010

Epoch 28/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 20s 849ms/step - loss: 0.2693 - mae: 0.6468 - val_loss: 0.2302 - val_mae: 0.6084 - learning_rate: 0.0010

Epoch 29/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 18s 649ms/step - loss: 0.2294 - mae: 0.5906 - val_loss: 0.2195 - val_mae: 0.5832 - learning_rate: 5.0000e-04

Epoch 30/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 12s 763ms/step - loss: 0.2140 - mae: 0.5521 - val_loss: 0.2144 - val_mae: 0.5630 - learning_rate: 5.0000e-04

Epoch 31/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 20s 772ms/step - loss: 0.2879 - mae: 0.6550 - val_loss: 0.2140 - val_mae: 0.5604 - learning_rate: 5.0000e-04

Epoch 32/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 21s 848ms/step - loss: 0.1955 - mae: 0.5274 - val_loss: 0.2169 - val_mae: 0.5663 - learning_rate: 5.0000e-04

Epoch 33/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 18s 652ms/step - loss: 0.2071 - mae: 0.5200 - val_loss: 0.2178 - val_mae: 0.5703 - learning_rate: 5.0000e-04

Epoch 34/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 12s 850ms/step - loss: 0.1792 - mae: 0.4974 - val_loss: 0.2089 - val_mae: 0.5516 - learning_rate: 5.0000e-04

Epoch 35/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 18s 642ms/step - loss: 0.2178 - mae: 0.5735 - val_loss: 0.2024 - val_mae: 0.5318 - learning_rate: 5.0000e-04

Epoch 36/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 11s 819ms/step - loss: 0.2237 - mae: 0.5832 - val_loss: 0.2029 - val_mae: 0.5412 - learning_rate: 5.0000e-04

Epoch 37/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 12s 845ms/step - loss: 0.2214 - mae: 0.5517 - val_loss: 0.2025 - val_mae: 0.5359 - learning_rate: 5.0000e-04

Epoch 38/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 20s 846ms/step - loss: 0.1960 - mae: 0.5301 - val_loss: 0.2093 - val_mae: 0.5545 - learning_rate: 5.0000e-04

Epoch 39/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 12s 889ms/step - loss: 0.1941 - mae: 0.5460 - val_loss: 0.2100 - val_mae: 0.5498 - learning_rate: 2.5000e-04

Epoch 40/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 20s 854ms/step - loss: 0.1738 - mae: 0.4803 - val_loss: 0.2053 - val_mae: 0.5411 - learning_rate: 2.5000e-04

Fold 4 results:

Test MSE: 0.4319

Test MAE: 0.8072

Test R^2: -0.0795

Per-bin evaluation:

Bin 0 (n=21): MSE=0.6639, MAE=0.7444

Bin 1 (n=14): MSE=0.1554, MAE=0.3310

Bin 2 (n=3): MSE=1.4882, MAE=1.0944

Bin 3 (n=5): MSE=5.0620, MAE=2.2323

Predictions saved to /content/drive/MyDrive/test_predictions_regression_fold_4.csv

Sample predictions:

True: 0.00, Predicted: 0.70

True: 1.00, Predicted: 1.10

True: 1.00, Predicted: 0.48

True: 0.00, Predicted: 0.77

True: 0.00, Predicted: 0.93

True: 0.00, Predicted: 0.96

True: 0.00, Predicted: 1.01

True: 1.00, Predicted: 0.45

True: 3.00, Predicted: 0.46

True: 0.00, Predicted: 1.44

Processing fold 5/5

Sample weight distribution: (array([0.89218834, 1.44921525]), array([225, 54]))

Training samples: 279

Test samples: 43

Training label distribution (rounded): [75 54 75 75]

Test label distribution (rounded): [19 9 9 6]

Model architecture:

Model: "functional_9"

Layer (type)	Output Shape	Param #	Connected to
mfcc_input (InputLayer)	(None, 500, 101)	0	-

```
│ egemaps_input       │ (None, 500, 101) │          0 │ -                 │
│ (InputLayer)        │                  │            │                   │
├─────────────────────┼──────────────────┼────────────┼──────
──────────────┤
│ not_equal_12        │ (None, 500, 101) │          0 │ mfcc_input[0][0]  │
│ (NotEqual)          │                  │            │                   │
├─────────────────────┼──────────────────┼────────────┼──────
──────────────┤
│ not_equal_13        │ (None, 500, 101) │          0 │ egemaps_input[0]… │
│ (NotEqual)          │                  │            │                   │
├─────────────────────┼──────────────────┼────────────┼──────
──────────────┤
│ masking_12          │ (None, 500, 101) │          0 │ mfcc_input[0][0]  │
│ (Masking)           │                  │            │                   │
├─────────────────────┼──────────────────┼────────────┼──────
──────────────┤
│ any_12 (Any)        │ (None, 500)      │          0 │ not_equal_12[0][… │
├─────────────────────┼──────────────────┼────────────┼──────
──────────────┤
│ masking_13          │ (None, 500, 101) │          0 │ egemaps_input[0]… │
│ (Masking)           │                  │            │                   │
├─────────────────────┼──────────────────┼────────────┼──────
──────────────┤
│ any_13 (Any)        │ (None, 500)      │          0 │ not_equal_13[0][… │
├─────────────────────┼──────────────────┼────────────┼──────
──────────────┤
│ lstm_26 (LSTM)      │ (None, 64)       │     42,496 │ masking_12[0][0], │
```

```
│                    │                   │            │ any_12[0][0]      │
├────────────────────┼───────────────────┼────────────┼───────────────────
──────────────────┤
│ lstm_27 (LSTM)     │ (None, 64)        │     42,496 │ masking_13[0][0], │
│                    │                   │            │ any_13[0][0]      │
├────────────────────┼───────────────────┼────────────┼───────────────────
──────────────────┤
│ batch_normalizatio… │ (None, 64)        │        256 │ lstm_26[0][0]     │
│ (BatchNormalizatio… │                   │            │                   │
├────────────────────┼───────────────────┼────────────┼───────────────────
──────────────────┤
│ batch_normalizatio… │ (None, 64)        │        256 │ lstm_27[0][0]     │
│ (BatchNormalizatio… │                   │            │                   │
├────────────────────┼───────────────────┼────────────┼───────────────────
──────────────────┤
│ dropout_20         │ (None, 64)        │          0 │ batch_normalizat… │
│ (Dropout)          │                   │            │                   │
├────────────────────┼───────────────────┼────────────┼───────────────────
──────────────────┤
│ dropout_21         │ (None, 64)        │          0 │ batch_normalizat… │
│ (Dropout)          │                   │            │                   │
├────────────────────┼───────────────────┼────────────┼───────────────────
──────────────────┤
│ concatenate_9      │ (None, 128)       │          0 │ dropout_20[0][0], │
│ (Concatenate)      │                   │            │ dropout_21[0][0]  │
├────────────────────┼───────────────────┼────────────┼───────────────────
──────────────────┤
```

```
│ dense_14 (Dense)    │ (None, 32)        │      4,128 │ concatenate_9[0]… │
├─────────────────────┼───────────────────┼────────────┼─────────
──────────────────┤
│ dropout_22          │ (None, 32)        │          0 │ dense_14[0][0]    │
│ (Dropout)           │                   │            │                   │
├─────────────────────┼───────────────────┼────────────┼─────────
──────────────────┤
│ dense_15 (Dense)    │ (None, 1)         │         33 │ dropout_22[0][0]  │
└─────────────────────┴───────────────────┴────────────┴─────────
──────────────────┘

 Total params: 89,665 (350.25 KB)

 Trainable params: 89,409 (349.25 KB)

 Non-trainable params: 256 (1.00 KB)

Epoch 1/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 22s 917ms/step - loss: 1.8478 - mae: 2.3551 - val_loss: 0.8697 - val_mae: 1.3505 - learning_rate: 0.0010

Epoch 2/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 19s 795ms/step - loss: 1.1807 - mae: 1.6557 - val_loss: 0.7080 - val_mae: 1.1871 - learning_rate: 0.0010

Epoch 3/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 18s 630ms/step - loss: 0.9776 - mae: 1.4052 - val_loss: 0.5721 - val_mae: 1.0486 - learning_rate: 0.0010

Epoch 4/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 12s 729ms/step - loss: 0.7757 - mae: 1.2051 - val_loss: 0.5040 - val_mae: 0.9714 - learning_rate: 0.0010

Epoch 5/50
```

14/14 ━━━━━━━━━━━━━━━━━━━━ 21s 775ms/step - loss: 0.8212 - mae: 1.2413 - val_loss: 0.4812 - val_mae: 0.9548 - learning_rate: 0.0010

Epoch 6/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 22s 854ms/step - loss: 0.6523 - mae: 1.0835 - val_loss: 0.4306 - val_mae: 0.8795 - learning_rate: 0.0010

Epoch 7/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 11s 810ms/step - loss: 0.6175 - mae: 1.0038 - val_loss: 0.4148 - val_mae: 0.8483 - learning_rate: 0.0010

Epoch 8/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 21s 839ms/step - loss: 0.4727 - mae: 0.8645 - val_loss: 0.3653 - val_mae: 0.7916 - learning_rate: 0.0010

Epoch 9/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 11s 813ms/step - loss: 0.5886 - mae: 1.0198 - val_loss: 0.3153 - val_mae: 0.7406 - learning_rate: 0.0010

Epoch 10/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 21s 842ms/step - loss: 0.5323 - mae: 0.9171 - val_loss: 0.2933 - val_mae: 0.7031 - learning_rate: 0.0010

Epoch 11/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 11s 815ms/step - loss: 0.5711 - mae: 0.9564 - val_loss: 0.3236 - val_mae: 0.7232 - learning_rate: 0.0010

Epoch 12/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 9s 650ms/step - loss: 0.5153 - mae: 0.9080 - val_loss: 0.3253 - val_mae: 0.7058 - learning_rate: 0.0010

Epoch 13/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 13s 827ms/step - loss: 0.4678 - mae: 0.8908 - val_loss: 0.2722 - val_mae: 0.6387 - learning_rate: 0.0010

Epoch 14/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 18s 643ms/step - loss: 0.4950 - mae: 0.8999 - val_loss: 0.2541 - val_mae: 0.6361 - learning_rate: 0.0010

Epoch 15/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 12s 747ms/step - loss: 0.4302 - mae: 0.8204 - val_loss: 0.2641 - val_mae: 0.6625 - learning_rate: 0.0010

Epoch 16/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 12s 843ms/step - loss: 0.3624 - mae: 0.7353 - val_loss: 0.2376 - val_mae: 0.6111 - learning_rate: 0.0010

Epoch 17/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 9s 646ms/step - loss: 0.4683 - mae: 0.8461 - val_loss: 0.2217 - val_mae: 0.5564 - learning_rate: 0.0010

Epoch 18/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 11s 681ms/step - loss: 0.3781 - mae: 0.7497 - val_loss: 0.2091 - val_mae: 0.5332 - learning_rate: 0.0010

Epoch 19/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 12s 839ms/step - loss: 0.4022 - mae: 0.7892 - val_loss: 0.2278 - val_mae: 0.5646 - learning_rate: 0.0010

Epoch 20/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 19s 706ms/step - loss: 0.3648 - mae: 0.7338 - val_loss: 0.2297 - val_mae: 0.5779 - learning_rate: 0.0010

Epoch 21/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 12s 852ms/step - loss: 0.3667 - mae: 0.7454 - val_loss: 0.2201 - val_mae: 0.5708 - learning_rate: 0.0010

Epoch 22/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 10s 734ms/step - loss: 0.3462 - mae: 0.7297 - val_loss: 0.2114 - val_mae: 0.5649 - learning_rate: 5.0000e-04

Epoch 23/50

```
14/14 ━━━━━━━━━━━━━━━━━━━━ 10s 687ms/step - loss: 0.2845 - mae: 0.6280 - val_loss: 0.2058 - val_mae: 0.5610 - learning_rate: 5.0000e-04
Epoch 24/50
14/14 ━━━━━━━━━━━━━━━━━━━━ 12s 836ms/step - loss: 0.2852 - mae: 0.6330 - val_loss: 0.1955 - val_mae: 0.5391 - learning_rate: 5.0000e-04
Epoch 25/50
14/14 ━━━━━━━━━━━━━━━━━━━━ 19s 700ms/step - loss: 0.3111 - mae: 0.6532 - val_loss: 0.2036 - val_mae: 0.5357 - learning_rate: 5.0000e-04
Epoch 26/50
14/14 ━━━━━━━━━━━━━━━━━━━━ 12s 845ms/step - loss: 0.3371 - mae: 0.6997 - val_loss: 0.1877 - val_mae: 0.5132 - learning_rate: 5.0000e-04
Epoch 27/50
14/14 ━━━━━━━━━━━━━━━━━━━━ 19s 702ms/step - loss: 0.3177 - mae: 0.6626 - val_loss: 0.1879 - val_mae: 0.5132 - learning_rate: 5.0000e-04
Epoch 28/50
14/14 ━━━━━━━━━━━━━━━━━━━━ 21s 781ms/step - loss: 0.2448 - mae: 0.5729 - val_loss: 0.1876 - val_mae: 0.5080 - learning_rate: 5.0000e-04
Epoch 29/50
14/14 ━━━━━━━━━━━━━━━━━━━━ 21s 841ms/step - loss: 0.3127 - mae: 0.6874 - val_loss: 0.1912 - val_mae: 0.5195 - learning_rate: 5.0000e-04
Epoch 30/50
14/14 ━━━━━━━━━━━━━━━━━━━━ 18s 636ms/step - loss: 0.2793 - mae: 0.6290 - val_loss: 0.1845 - val_mae: 0.4970 - learning_rate: 5.0000e-04
Epoch 31/50
14/14 ━━━━━━━━━━━━━━━━━━━━ 13s 790ms/step - loss: 0.2656 - mae: 0.6041 - val_loss: 0.1760 - val_mae: 0.4700 - learning_rate: 5.0000e-04
Epoch 32/50
```

14/14 ━━━━━━━━━━━━━━━━━━━━ 18s 643ms/step - loss: 0.3194 - mae: 0.6787 - val_loss: 0.1745 - val_mae: 0.4535 - learning_rate: 5.0000e-04

Epoch 33/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 11s 794ms/step - loss: 0.2850 - mae: 0.6356 - val_loss: 0.1707 - val_mae: 0.4515 - learning_rate: 5.0000e-04

Epoch 34/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 18s 654ms/step - loss: 0.2413 - mae: 0.5909 - val_loss: 0.1702 - val_mae: 0.4688 - learning_rate: 5.0000e-04

Epoch 35/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 11s 773ms/step - loss: 0.2649 - mae: 0.6013 - val_loss: 0.1742 - val_mae: 0.4759 - learning_rate: 5.0000e-04

Epoch 36/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 12s 845ms/step - loss: 0.2683 - mae: 0.6238 - val_loss: 0.1727 - val_mae: 0.4540 - learning_rate: 5.0000e-04

Epoch 37/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 20s 766ms/step - loss: 0.2391 - mae: 0.5546 - val_loss: 0.1710 - val_mae: 0.4597 - learning_rate: 5.0000e-04

Epoch 38/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 20s 728ms/step - loss: 0.2627 - mae: 0.6135 - val_loss: 0.1688 - val_mae: 0.4540 - learning_rate: 2.5000e-04

Epoch 39/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 10s 698ms/step - loss: 0.2545 - mae: 0.6044 - val_loss: 0.1666 - val_mae: 0.4418 - learning_rate: 2.5000e-04

Epoch 40/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 12s 846ms/step - loss: 0.2448 - mae: 0.5721 - val_loss: 0.1635 - val_mae: 0.4372 - learning_rate: 2.5000e-04

Epoch 41/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 10s 746ms/step - loss: 0.2293 - mae: 0.5760 - val_loss: 0.1625 - val_mae: 0.4335 - learning_rate: 2.5000e-04

Epoch 42/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 22s 825ms/step - loss: 0.2375 - mae: 0.5575 - val_loss: 0.1656 - val_mae: 0.4400 - learning_rate: 2.5000e-04

Epoch 43/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 18s 643ms/step - loss: 0.2010 - mae: 0.5235 - val_loss: 0.1673 - val_mae: 0.4338 - learning_rate: 2.5000e-04

Epoch 44/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 12s 845ms/step - loss: 0.2026 - mae: 0.5280 - val_loss: 0.1678 - val_mae: 0.4423 - learning_rate: 2.5000e-04

Epoch 45/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 18s 640ms/step - loss: 0.2124 - mae: 0.5438 - val_loss: 0.1683 - val_mae: 0.4471 - learning_rate: 1.2500e-04

Epoch 46/50

14/14 ━━━━━━━━━━━━━━━━━━━━ 12s 843ms/step - loss: 0.2129 - mae: 0.5145 - val_loss: 0.1654 - val_mae: 0.4375 - learning_rate: 1.2500e-04

Fold 5 results:

Test MSE: 0.6295

Test MAE: 1.0614

Test R^2: -0.3249

Per-bin evaluation:

Bin 0 (n=19): MSE=0.9285, MAE=0.8577

Bin 1 (n=9): MSE=0.3795, MAE=0.5268

Bin 2 (n=9): MSE=1.5456, MAE=1.1638

Bin 3 (n=6): MSE=5.6341, MAE=2.3548

Predictions saved to /content/drive/MyDrive/test_predictions_regression_fold_5.csv

Sample predictions:

True: 2.00, Predicted: 1.25

True: 0.00, Predicted: 1.35

True: 0.00, Predicted: 0.61

True: 2.00, Predicted: 0.51

True: 1.00, Predicted: 0.28

True: 3.00, Predicted: 0.47

True: 3.00, Predicted: 1.27

True: 2.00, Predicted: 1.00

True: 3.00, Predicted: 0.73

True: 0.00, Predicted: 0.32

Cross-validation results: MSE=0.5468±0.0870, MAE=0.9492±0.0993, R^2=-0.3624±0.2038

Pipeline execution completed!

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